pmsimstats-ng Documentation Compendium

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Purpose. This compendium consolidates the novel technical content from all source documents in `~/Dropbox/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/`. It unifies mathematical derivations, empirical results, audit findings, and implementation guidance that is currently scattered across twenty-six numbered .md and .tex files. Each section cites the source documents that contributed to it. Sections or claims that have been superseded by subsequent development are flagged explicitly.

The documentation corpus was last substantially updated on 2026-04-16 (audit report) and 2026-04-15 (parity workflow, revision response to reviewer on document 02). The current repository state reflects the 2026-04-08 merge that reorganized code into `implementations/`, added Architecture A (mean moderation) alongside Architecture B (multivariate normal differential correlation), and moved the pre-merge tidyverse pipeline (`analysis/2025/`) to an external archive.

Review note (2026-10-07). This compendium records the state of the project in April 2026, and several of its statements have since been corrected or superseded. Outright errors (wrong numbers, wrong mathematics, wrong references, notation) were fixed in place on 2026-10-07. Superseded statements are kept as written, each with a short dated note pointing to the newer document. The main ones: the positive-definiteness ceilings on c_{bm} (Sections 2.1, 2.7 and 3.3; see `docs/34`, `docs/35` and `docs/36`); the account of the published carryover collapse (Sections 1.5 and 5.7; see `docs/36` and `docs/37`); the headline 40-60 percent versus 10-15 percent power-loss contrast (Sections 1.4, 16.1 and 18; see paper 01 and `docs/37`); and the recommended analysis model (Section 4; see `docs/45`, `docs/46` and `docs/49`). The data-generating features that a complete N-of-1 DGP needs are reviewed in `docs/51`.

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Notation index and glossary

The notation follows the canonical table in `analysis/report/NOTATION.md`. A symbol marked *canonical* has the meaning given there. A symbol marked *local* is used only in this compendium and was chosen so that it does not collide with a canonical symbol. Code identifiers (`c.bm`, `Dbc`, `Sx`, `tsd`, `tod`, `lambda_cor`, `carryover_t1half`) appear only in code spans. The package passes the moderation strength as `c.bm` under both architectures. In the mathematics it is written c_{bm} under covariance moderation and β_{bm} under mean moderation. Where a table reports both architectures at one value, the shared label is $c_{bm} = \beta_{bm}$ (NOTATION rule 2). N is the total sample size across randomization paths. However, the April 2026 runs quoted here predate the 2026-08 driver audit, so whether a quoted N is a total or a per-path count has not been re-verified (NOTATION rule 4).

Table 1: Notation index: symbols used in this compendium

Symbol	Meaning	Status
i, t	participant and timepoint (weeks)	canonical
N	total number of participants	canonical
Y_{it}	observed symptom score (code `Sx`)	canonical
BR_{it}, PB_{it}, TV_{it}	biologic-response, placebo-belief and time-variant components	canonical
ε_{it}, u_i	residual and participant random intercept	canonical
D_{it}	binary drug state	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`)	canonical
t_{sd}, t_{od}	time since discontinuation, time on drug (code `tsd`, `tod`)	canonical
$t_{1/2}$	carryover half-life, weeks (code `carryover_t1half`)	canonical
λ	decay rate $\ln 2 / t_{1/2}$ (code `lambda_cor`)	canonical
B_i	raw biomarker (code `bm`)	canonical
b_i	standardized biomarker $(B_i - \mu_{bm})/\sigma_{bm}$ (code `bm_z`)	canonical

Symbol	Meaning	Status
μ_{bm}, σ_{bm}	biomarker mean and SD (124.33, 15.36)	canonical (σ_{bm}); local (μ_{bm})
σ_{BR}	SD of BR (8)	canonical
$\sigma_{PB,t}$	SD of PB at visit t, scaled by expectancy	local
η_t	expectancy at visit t: 1 open label, 0.5 blinded, 0 baseline (code `e`)	canonical
c_{bm}	covariance-moderation strength: on-drug correlation of B_i with BR_{it}	canonical
β_{bm}	mean-moderation multiplier	canonical
$\beta_{bm:D}$	biomarker-by-treatment interaction coefficient (`bm:Dbc`)	canonical
β_D	treatment main effect	canonical
$\beta_0, \ldots, \beta_3$	other fixed effects of the model being described	local
ρ	AR(1) within-factor correlation per week (nlme calls the `corCAR1` parameter Phi)	canonical
ϕ_t	decay factor $2^{-t_{sd}/t_{1/2}}$ off drug, 0 before first exposure	local (as in `docs/37`)
g_t	coupling: the fraction of c_{bm} present in $\mathrm{Cor}(B_i, BR_{it})$	local (as in `docs/37`)
r_t	biomarker-response correlation at visit t, $c_{bm} g_t$	local (as in `docs/37`)
h_t	moderation profile of a mean-moderation shift (Section 8.1, Attempt 2)	local (as in `docs/37`)
s	carryover scale factor in the analysis indicator (code `carryover_scalefactor`, default 1)	local
$\sigma_{BR,t}$	a hypothetical time-varying SD of BR (Section 5.8)	local

Symbol	Meaning	Status
$\bar{Y}_{i,\text{on}}$, $\bar{Y}_{i,\text{off}}$, Δ_i, σ_Δ	participant phase means, their difference, and its SD (Sections 10, 13)	local
T, K, L	visits per participant, number and length of drug blocks (Section 13)	local
$\mu_{BR,t}$	mean of BR at visit t	local
Δ_{on}, Δ_{off}	spread of expected BR between participants at ± 1 SD of the biomarker	local
σ_u^2, σ_ε^2, σ_T^2	random-intercept, residual and total within-participant variance (Section 4.3)	local
ω_0, ω_1	intercept and biomarker slope of a change-score regression (Sections 10, 13)	local
ξ_0, ξ_1, e_i	intercept, slope and error of the second-stage regression (Section 10)	local
$\pi(B_i)$	latent-class membership probability (Section 11; paper 03 usage)	local
Z_i	latent class label	canonical
$\text{cond}(\Sigma)$	condition number of a covariance matrix	local
α	nominal significance level	canonical

Glossary.

- **Architecture A / Architecture B.** Former labels for the mean-moderation architecture (`dgp_architecture = 'mean_moderation'`) and the covariance-moderation architecture (`'mvn'`). The labels are kept in this historical document; NOTATION.md now names the architectures by mechanism.
- **N-of-1 (design).** The label these April 2026 sources use for the Hendrickson (2020) Hybrid design (OL, then BDC, then a brief crossover). It is called Hybrid elsewhere in the project.
- **OL, OL+BDC, CO.** Open-label, open-label followed by blinded discontinuation, and crossover designs.

- **ER.** An older name for the expectancy (placebo-belief) component, now PB; it has been renamed PB throughout this document.
- **Step, proportional and graded coupling.** Rules for the off-drug biomarker-response correlation (Section 5). The current package uses graded coupling, $g_t = \phi_t$ off drug and $g_t = 0$ before first exposure (inspected, `R/generateData.R`, 2026-10-07).
- **PD.** Positive definite (of a covariance matrix); the *ceiling* is the largest c_{bm} for which the matrix is PD.
- **E1/E2/E3.** The project's current labels for the three carryover analysis specifications (NOTATION Part 3). Section 12 predates them.

0. Source Document Inventory and Antiquation Map

The source files under `docs/` are listed below with their role and their standing as of 2026-04-21. Files marked 'Current' are faithful to the code. Files marked 'Historical' document decisions or states that have been implemented; their technical content is preserved here because the reasoning remains instructive. Files marked 'Superseded' describe methodology that the revised code no longer uses; readers should consult this compendium or the currently-labeled source rather than those files.

Table 2: Source documents with their role and current standing

File	Role	Standing
00-documentation-index.tex	Navigation index	Current
01-codebase-overview.tex	Package structure and parameter table	Current
02-revision-response-latent-class-expanded.md	Psychometric connection	Current
02-revision-response-to-reviewer.md	Editorial revision log	Current
03-audit-and-revision-report.tex	Seven DGP corrections	Current (foundational)
04-revised-power-analysis.tex	Validated power estimates	Current
06-ar1-residual-correlation.tex	nlme plus corCAR1 transition	Current
07-positive-definiteness-failures.tex	PD failure analysis	Current
08-biomarker-correlation-decay.tex	lambda_cor derivation	Current
09-carryover-correlation-artifact.tex	Step-function artifact	Current (diagnostic)

File	Role	Standing
10-carryover-analysis-model-assessment.tex	2x2 factorial plan	Current (proposed study)
13-figure4-walkthrough.tex	Publication methodology	Superseded
14-trial-analysis-implementation-guide.tex	Collaborator deliverables	Current
15-tidyverse-alignment-plan.tex	Pre-merge port plan	Historical
16-tidyverse-code-review.tex	Pre-merge code review	Historical
17-simplification-plan.tex	Alternative simplification	Historical (not adopted)
18-response-parameter-sensitivity.tex	Figure 5 validation	Current
19-mean-moderation-implementation-notes.tex	Architecture A derivation	Current
19-mean-moderation-implementation-notes.md	Quick-reference summary	Current
20-annotated-bibliography.tex	Literature synthesis	Current reference
21-architecture-comparison-summary.md	Empirical comparison	Current
22-journal-target-recommendation.md	Submission strategy	Current
22-tidyverse-development-parity-workflow.md	Parity workflow	Current
audit-2026-04-16.md	Technical audit	Current (most recent)
drafting-02-rgt/working.tex	Working draft of doc 02	Working file (no material differences)
07a-pd-constraints-theory.tex	PD constraints derivation (Sylvester, Gershgorin, correlation hierarchy)	Current (migrated 2026-04-23 from pmsimstats2025)
09a-carryover-bias-variance.md	Non-monotonic power under model misspecification (bias-variance tradeoff)	Current (migrated 2026-04-23 from pmsimstats2025)

File	Role	Standing
24-component-decomposition-pedagogy.md	BR/ER/TV conceptual guide with prazosin-PTSD narrative	Current (migrated 2026-04-23 from pmsimstats2025)
archive/correlation-structure-alternatives.md	Alternative correlation structures (Matern, rational-quadratic, power-law)	Historical (migrated 2026-04-23; exploratory)
archive/hendrickson-pseudocode-comparison.md	Side-by-side pseudocode audit trail vs. Hendrickson original	Historical (migrated 2026-04-23)

Note (2026-10-07). The standings above are as of 2026-04-21. Several listed files are no longer in `docs/`: the two `02-revision-response-*.md` files, `10-carryover-analysis-model-assessment.tex`, `audit-2026-04-16.md` and `drafting-02-rgt/`. Documents `docs/24` to `docs/51` postdate this compendium. Of these, `docs/34` to `docs/37`, `docs/45`, `docs/46`, `docs/49` and `docs/51` supersede parts of Sections 1 to 5 (see the dated notes below).

Four key antiquation points to highlight:

1. **Doc 13 (Figure 4 Walkthrough)** describes the publication methodology (compound symmetry, `lme4::lmer`, step-function BM-BR correlation, carryover half-lives of 0.1 and 0.2 weeks). The revised code no longer uses any of these elements. The document is retained as a record of the baseline from which the seven DGP corrections (Section 2) were derived.

2. **Doc 15 (Tidyverse Alignment Plan)** describes a pre-merge roadmap for porting `analysis/2025/pm_functions.R` to parity with the original data.table code. The merge on 2026-04-08 executed most of this plan, moved the pre-merge tidyverse file to an external archive, and reorganized the code into `implementations/`. Function names and paths referenced in doc 15 no longer resolve.

3. **Doc 16 (Tidyverse Code Review)** reviews the same pre-merge file. Its findings informed the 2026-04-08 rewrite. Line numbers and function names in doc 16 refer to the archived file, not the current `implementations/tidyverse/R/functions.R`.

4. **Doc 17 (Simplification Plan)** proposed a substantially simpler framework (random intercept with independent errors, ANCOVA on phase-mean change scores, reduced parameter space). The merge took a different path: it added Architecture A rather than simplifying. The document is retained as a record of considered alternatives and contains the most

thorough formal specification of six alternative analysis methods in the project.

One execution-blocking issue was carried forward at the time of writing (audit-2026-04-16, finding `A1`): `analysis/carryover_factorial/02-run-factorial-2025.R:20` sources `analysis/2025/01-pm-functions.R`, which has been moved to an external archive. The script fails at load time until the source target is updated to `implementations/tidyverse/R/functions.R` or a copy from the archive is restored.

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1. The Dual Data-Generating Process Architectures

Source documents: 02-dgp-mean-moderation-vs-mvn.tex; 19-mean-moderation-implementation-notes.tex; 21-architecture-comparison-summary.md; 02-revision-response-latent-class-expanded.md.

1.1 Architecture A: Direct Mean Moderation

The biomarker enters the mean structure of the outcome as a multiplicative scaling of the treatment effect:

$$Y_{it} = \beta_0 + \beta_D \cdot D_{it} \cdot (1 + \beta_{bm} \cdot b_i) + \varepsilon_{it}$$

Expanding the product gives the additive interaction model

$$Y_{it} = \beta_0 + \beta_D D_{it} + \beta_2 b_i + \beta_{bm:D} D_{it} b_i + \varepsilon_{it}$$

with $\beta_2 = 0$ and $\beta_{bm:D} = \beta_D \beta_{bm}$. The multiplicative form is therefore the special case of the additive model that has no biomarker main effect. The identity is algebraic and does not depend on centering. Centering (b_i rather than B_i) only makes β_D the treatment effect at the mean biomarker. (Corrected 2026-10-07: the earlier text said that the two forms are equivalent under centering.)

Properties:

- The interaction is deterministic given biomarker and treatment status. This is a population-level mean-structure claim, not a statement about individual outcomes.
- β_{bm} is a population parameter, uniform across participants.
- The signal lives in the first moment of the outcome distribution.

- Under carryover, the ratio of drug effect between high- and low-biomarker participants is preserved. Power loss is modest.

Implementation in the pmsimstats-ng framework: after an MVN draw from a covariance matrix that omits the BM-BR block, each participant's BR factor at on-drug timepoints is shifted by a standardized biomarker deviation:

$$BR_{it}^{\text{mod}} = BR_{it}^{\text{base}} + \beta_{bm}\, b_i\, \mathbf{1}[D_{it} = 1]\, \sigma_{BR}, \qquad b_i = (B_i - \mu_{bm})/\sigma_{bm}$$

The shift per one standard deviation of biomarker is $\beta_{bm}\sigma_{BR}$. For $\beta_{bm} = 0.45$ and $\sigma_{BR} = 8$ (the value in `extracted_rp`, verified) this is 3.6 BR units, which matches the conditional expectation under Architecture B at $c_{bm} = 0.45$. In code the multiplier is passed as `c.bm`. *Note (2026-10-07):* `generateData()` also offers `moderation_scaling = 'trajectory'`. Under that option the shift follows the on-drug BR mean curve, normalized so that its mean on-drug magnitude is σ_{BR} (inspected, `R/generateData.R`).

1.2 Architecture B: Multivariate Normal Differential Correlation

The biomarker and the biologic response factor are jointly multivariate normal, with a treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_{bm} \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \\ c_{bm} \cdot \exp(-\lambda \cdot t_{sd}) & \text{if } D_{it} = 0,\ t_{sd} > 0 \\ 0 & \text{before first exposure} \end{cases}$$

The zero before first exposure (for example, the placebo-first CO path) is the current package behavior (inspected, `buildSigma()`); the earlier text omitted that case.

The conditional expectation on drug is:

$$E[BR_{it} \mid B_i, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_{bm}} \cdot (B_i - \mu_{bm}) = \mu_{BR}(t) + c_{bm}\, \sigma_{BR}\, b_i$$

Off drug, with carryover half-life $t_{1/2}$ and $\lambda = \ln 2 / t_{1/2}$:

$$E[BR_{it} \mid B_i, D_{it} = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \sigma_{BR}\, b_i$$

Properties:

- The interaction is probabilistic: a tendency, not a deterministic scaling.
- c_{bm} is a correlation coefficient with $|c_{bm}| \leq 1$.
- The signal lives in the second moment (the covariance structure).
- Carryover erodes the differential correlation. Power loss is substantial.

1.3 Mathematical Comparison

Architecture A yields an expected difference that scales cleanly with drug exposure:

$$E[Y_{it} \mid b_i, D_{it}] - E[Y_{jt} \mid b_j, D_{jt}] = \beta_D \cdot \beta_{bm} \cdot (b_i - b_j) \cdot D$$

for two participants i and j in the same drug state $D_{it} = D_{jt} = D$. The ratio of effects between high- and low-biomarker participants is preserved as D varies. If D is replaced by a carryover-decayed continuous indicator, the difference scales by the same factor, but the high/low ratio is unchanged.

Architecture B yields an on-drug difference:

$$E[BR_{it} \mid b_i, D = 1] - E[BR_{jt} \mid b_j, D = 1] = c_{bm} \cdot \sigma_{BR} \cdot (b_i - b_j)$$

Off drug, the difference is multiplied by $e^{-\lambda t_{sd}}$. The biomarker-specific signal erodes exponentially during washout.

1.4 Empirical Power Comparison

Setup (doc 21): $N = 70$, $c_{bm} = \beta_{bm} = 0.45$, `nlme::lme` plus `corCAR1` with the `bm:Dbc` interaction, 50 replicates. Half-lives in weeks.

Architecture B (MVN differential correlation):

Table 3: Architecture B power by design and carryover half-life, 50 replicates

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
N-of-1	0.82	0.64	0.50
CO	0.40	0.40	0.32
OL+BDC	0.62	0.38	0.24

Relative power loss at $t_{1/2} = 1.0$: N-of-1 39 percent, CO 20 percent, OL+BDC 61 percent.

Architecture A (mean moderation):

Table 4: Architecture A power by design and carryover half-life, 50 replicates

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
N-of-1	0.74	0.72	0.68
CO	0.38	0.36	0.34
OL+BDC	0.62	0.58	0.54

Relative power loss at $t_{1/2} = 1.0$: N-of-1 8 percent, CO 11 percent, OL+BDC 13 percent.

Monte Carlo standard error at power near 0.5 with 50 replicates is approximately $\sqrt{0.5 \cdot 0.5/50} = 0.07$.

Note (2026-10-07). `docs/19` (both the `.md` and `.tex` versions) reports the N-of-1 Architecture A row as 0.74 / 0.68 / 0.72, and doc 21 reports it as 0.74 / 0.72 / 0.68. The middle and last cells are swapped, and the source run was not located, so the discrepancy is unresolved. Either version lies within Monte Carlo error of flat power. These are 50-replicate pilot estimates. With per-cell MCSE of about 0.07, the relative losses above are imprecise to about ± 15 percentage points, and they should not be cited as the project's result. Paper 01 later re-ran the comparison at a matched total of $N = 70$. In that run Hybrid carryover losses were 6.9 percent under mean moderation and 15.4 percent under covariance moderation (as recorded in `analysis/report/NOTATION.md`, Part 2; inspected, not recomputed). `docs/37` gives the closed-form dependence of power on the half-life.

1.5 Why the Architectures Diverge

Two mechanisms operate simultaneously under Architecture B:

1. **Mean blurring** (shared with Architecture A). Off-drug BR means are inflated by residual carryover, compressing the on-drug versus off-drug treatment contrast. Both architectures experience this.

2. **Correlation erosion** (unique to Architecture B). The BM-BR correlation decays exponentially with t_{sd}. Because the interaction signal lives in the second moment, this directly weakens the signal. This mechanism has no analogue in Architecture A, where the interaction lives in the first moment.

Architecture A's interaction coefficient sees reduced `Dbc` contrast under carryover, but the signal-to-noise ratio degrades only modestly because the biomarker-by-drug signal scales proportionally with the reduced contrast. Architecture B's interaction coefficient is attacked from both sides: the treatment contrast in `Dbc` is compressed, and the biomarker-BR correlation generating the covariance signal is simultaneously decaying.

Note (2026-10-07). `docs/37` (derived and checked by simulation) qualifies this two-mechanism account. For the paired-difference statistic, carryover in the response *means* is absorbed by the contrast's intercept. The whole effect of the half-life on power then runs through the off-drug coupling $\bar{g}_{\text{off}}$, that is, through correlation erosion alone. Under mean moderation with an on-drug-only shift, power does not depend on the half-life (`docs/37`, Section 1, points 1 and 4). Mean blurring therefore explains little of the architecture contrast.

1.6 Biological Rationale and Architecture Selection

The choice between architectures is a claim about biology, not statistical convenience.

Architecture A is appropriate when the biomarker governs the **magnitude of the drug's biological effect** for each individual. Examples:

- Renal clearance (eGFR determines drug elimination).
- Receptor density moderation (target availability determines pharmacodynamics).
- CYP2D6 metabolizer status.
- Dose-response with biomarker-dependent effective dose.

Architecture B is appropriate when the biomarker predicts which participants are likely to respond, through shared variance with a drug-responsive physiological component that decays as the drug is eliminated. Examples:

- Blood pressure as PTSD subtype marker (elevated resting SBP indexes heightened central noradrenergic tone, a subtype preferentially responsive to prazosin as alpha-adrenergic antagonist).
- Baseline disease severity as treatment predictor.
- Inflammatory biomarkers in psychiatry.
- Genetic risk scores.

Decision table (doc 02):

Table 5: Decision criteria for choosing between Architecture A and Architecture B

Criterion	Architecture A	Architecture B
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact	Modest (10-15 pct)	Substantial (40-60 pct)

Criterion	Architecture A	Architecture B
Appropriate when	Biomarker determines effective dose or PK	Biomarker indexes latent subtype

The carryover-impact row restates the 50-replicate pilot of Section 1.4. The note there gives the later matched-N figures.

Reporting recommendation: simulation studies should explicitly state whether the biomarker-treatment interaction is mean moderation or differential correlation, provide the biological rationale, confirm that the carryover model operates consistently with the chosen architecture, and include sensitivity analyses under the alternative architecture when the biological mechanism is uncertain. When uncertain, Architecture B produces more conservative (lower) power estimates under carryover and is therefore the safer default for trial design.

1.7 Prevalence of Each Architecture in the Literature

Architecture A is the near-universal standard in clinical trial simulation, appearing in biomarker-stratified designs (Simon 2010), adaptive enrichment trials (Freidlin and Korn 2014), basket and umbrella trials (Renfro and Sargent 2017), platform trials, and biomarker-by-treatment interaction studies (Haller and Ulm 2018). It is standard in N-of-1 simulation (Zucker et al. 1997; Araujo et al. 2016; Duan et al. 2013) and in N-of-1 methodology on power and design (Wang and Schork 2019; Schork 2022).

Architecture B (MVN differential correlation) appears primarily in latent variable and structural equation modeling, Bayesian joint modeling, and copula-based simulation. The Hendrickson et al. (2020) framework appears to be unique in the N-of-1 trial literature in using differential correlation as the interaction mechanism.

Implication: published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through an Architecture B mechanism, because those studies do not account for the additional power loss that carryover produces when the interaction signal resides in the covariance structure rather than the mean structure.

---

2. The Seven DGP Corrections

Source document: 03-audit-and-revision-report.tex (dated 2026-03-24).

The revised codebase differs from the Hendrickson et al. (2020) publication code in seven substantive ways. Each is documented below with the mathematical form, the reason for the change, and the effect on inference.

2.1 Correction 1: AR(1) Autocorrelation

Replaces compound symmetry within each response factor.

$$\text{Cor}(X_{t_i}, X_{t_j}) = \rho^{|t_i - t_j|}$$

Under compound symmetry the correlation is constant regardless of time gap. Under AR(1) it decays exponentially with temporal separation, which is clinically more realistic. The change widens the PD-feasible range for the BM-BR correlation. *Corrected 2026-10-07* (the earlier text mixed two values of ρ in one comparison and overstated the AR(1) ceiling). Under the published compound-symmetry specification ($\rho = 0.8$, no carryover) the ceiling is 0.256 (`docs/36`, verified there). At the same $\rho = 0.8$, the separable AR(1) structure raises the Hybrid ceiling to 0.440 (`docs/36`). At the package value $\rho = 0.7$ and no carryover, the separable AR(1) ceiling is 0.474 in OL+BDC and 0.480 in Hybrid (`docs/35`), and the current `covar` AR(1) structure gives 0.452 and 0.456 (`docs/34`).

2.2 Correction 2: Exponential BM-BR Correlation Decay

Replaces the step-function rule under which the BM-BR correlation dropped to zero the moment a participant was off drug regardless of carryover residual.

$$\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot e^{-\lambda t_{sd}}$$

with $\lambda = \ln 2 / t_{1/2}$ by default (code `lambda_cor`). The correlation decays in proportion to the drug's pharmacokinetic elimination rate, and it is zero at off-drug visits before first exposure. The derivation from first principles appears in Section 5 below. `docs/37` shows that this graded coupling removes the published power collapse (see the note in Section 5.7).

2.3 Correction 3: nlme::lme with corCAR1

Replaces `lme4::lmer` with random intercept only.

```
nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc,
          random = ~1|ptID,
          correlation = nlme::corCAR1(form = ~t|ptID),
          data = datamerged,
          control = nlme::lmeControl(opt = 'optim',
                                     maxIter = 200,
                                     msMaxIter = 200))
```

Under the AR(1) DGP, `lmer` with random intercept alone inflates Type I error to 13-17 percent for CO and OL designs. The `corCAR1` residual structure restores nominal Type I control (Section 4 below).

2.4 Correction 4: Carryover Scale Factor Removal

Removes the undocumented scale factor of 2 that commit 8609f12 added to the carryover decay exponent:

- Original (commit 42ac030): `(1/2)^(t_sd / t_half)`.
- Ron Thomas edit (commit 8609f12): `(1/2)^(2 * t_sd / t_half)`, halving the effective half-life.
- Revised HEAD: `(1/2)^(t_sd / t_half)`. The parameter `t_half` means what it says.

2.5 Correction 5: Timepoint-1 Guard Removal

Commit 8609f12 introduced a guard `if (p > 1)` in the BM-BR correlation assignment loop that was intended to avoid an index error in the proportional-scaling formula. The guard inadvertently forced the BM-BR correlation at timepoint 1 to zero even when participants were on drug at that timepoint. The revised code iterates from `p = 1` without the guard.

2.6 Correction 6: Gompertz Origin-Passing

Replaces the naive parameterization `f(t) = M * exp(-d * exp(-r*t))`, which does not pass through the origin, with an offset-rescaled form that guarantees `f(0) = 0` and preserves the asymptote at `M`:

```
y <- maxr * exp(-disp * exp(-rate * t))
vert_offset <- maxr * exp(-disp)
y <- y - vert_offset
y <- y * (maxr / (maxr - vert_offset))
```

Doc 15 (tidyverse alignment plan) identifies this correction as the single largest source of divergence between the original and the pre-merge tidyverse implementation. The naive form shifts every component mean in the MVN distribution.

2.7 Correction 7: Updated Default Parameters

- Autocorrelation: 0.8 (compound symmetry) to 0.7 (AR(1)).

- Biomarker correlation: publication grid `{0, 0.3, 0.6}` reduced to `{0, 0.25, 0.45}`. PD-feasible maximum under AR(1) is 0.49 for OL; `c.bm = 0.6` fails PD checks for 7 of 9 trial paths at zero carryover and all 9 at higher carryover under Architecture B. *Note (2026-10-07):* these April 2026 grid-search figures were not reproduced by the later closed-form ceilings. At $\rho = 0.7$ the current `covar` structure gives design-level ceilings of 0.619 (CO), 0.456 (Hybrid) and 0.452 (OL+BDC) at $t_{1/2} = 0$. These rise with the half-life in Hybrid and OL+BDC, to 0.544 and 0.533 at $t_{1/2} = 1$ (`docs/34`, Table 4.1, verified there). On those figures CO paths admit $c_{bm} = 0.6$, so the statement that "all 9" paths fail at higher carryover is not supported. The reference value 0.45 lies within 0.002 to 0.006 of the Hybrid and OL+BDC ceiling without carryover.
- Carryover half-life: publication grid `{0, 0.1, 0.2}` weeks replaced with `{0, 0.5, 1.0}` weeks. The publication values are too short to be clinically meaningful and were used precisely where the step-function correlation artifact was largest.

2.8 Correction Status Across Implementations

From the 2026-04-16 audit:

Table 6: Status of the seven DGP corrections across the four implementations

Correction	original	original-extended	tidyverse	simple
1. AR(1) autocorrelation	yes	yes	yes	n/a
2. Exponential BM-BR decay	yes	yes	yes	n/a
3. nlme::lme with corCAR1	yes	yes	yes	no (OLS)
4. Scale factor removal	yes	yes	yes	n/a
5. Timepoint-1 guard removal	yes	yes	yes	n/a
6. Gompertz origin-passing	yes	yes	yes	partial
7. Updated defaults	yes	yes	yes	partial

All seven corrections are embodied in the three production implementations. The simple sandbox omits some by design (it uses ANCOVA on phase-mean change scores rather than LME).

---

3. Positive Definiteness of the Covariance Matrix

Source documents: 07-positive-definiteness-failures.tex; 03-audit-and-revision-report.tex; 04-revised-power-analysis.tex.

3.1 The Problem

The DGP builds a 26-by-26 covariance matrix per trial path from user-specified correlations and standard deviations. When the result is not positive definite, `generateData()` projects to the nearest PD matrix via `corpcor::make.positive.definite(sigma, tol = 1e-3)`. In the publication code the correction was silent.

Failure rates by commit:

Table 7: Positive definiteness failure rates of covariance matrices by commit

Commit	Description	Failures	Rate
42ac030	Initial publication	40 of 162	24.7 pct
8609f12	Ron Thomas edits	102 of 162	63.0 pct
f6ee86d	Autocorr typo fix	102 of 162	63.0 pct
f70f86d	Carryover added	102 of 162	63.0 pct

All 162 matrices in the standard parameter sweep were ill-conditioned ($\text{cond}(\Sigma) > 100$) under compound symmetry.

Condition number statistics (all 162 matrices):

Table 8: Covariance matrix condition number statistics by commit across 162 matrices

Commit	Median cond(Σ)	$> 10^2$	$> 10^3$	$> 10^6$
42ac030	229	162/162	40/162	40/162
8609f12	3.1 times 10^18	162/162	102/162	102/162

Minimum eigenvalues of failed matrices ranged from -33.26 (worst) to -0.52 (least severe), on a dominant eigenvalue of approximately 700. The most extreme corrections displaced about 5 percent of the matrix structure, concentrated in eigenvectors affecting the BM-BR and cross-phase PB correlations disproportionately.

Note (2026-10-07). `docs/36`, working from the published commit `58b32a9`, reports 40 of 162 failing matrices. These are every matrix with $c_{bm} > 0$ and no carryover in CO, OL+BDC and Hybrid, plus the placebo-first CO path at every half-life. It attributes them to the requested biomarker-response correlation exceeding the CS ceiling of 0.256 to 0.262 rather than to the response block (`docs/36`, Section 1, point 1). This differs from the OL+BDC expectancy account in Section 3.2.

3.2 Root Cause: OL+BDC Expectancy Discontinuity

Failures were concentrated in the OL+BDC design (83-92 percent failure rate, 32 of 40 failures under the initial commit). The OL design never fails. The mechanism:

- Expectancy (code `e`) changes abruptly: $\eta_t = 1.0$ in the open-label phase (4 timepoints) and $\eta_t = 0.5$ in the blinded phase (4 timepoints).
- The PB standard deviation scales with expectancy: $\sigma_{PB,t} = \sigma_{PB}\,\eta_t$, producing the pattern $(10, 10, 10, 10, 5, 5, 5, 5)$.
- Cross-phase covariance under compound symmetry with $\rho = 0.8$: $0.8 \cdot 10 \cdot 5 = 40$, yielding a pairwise correlation of $40/\sqrt{100 \cdot 25} = 0.8$, consistent at the pair level.
- However, compound symmetry requires all same-factor timepoint pairs to have identical correlation regardless of temporal separation. With heterogeneous variances across phases this becomes self-contradictory: a single correlation coefficient cannot simultaneously satisfy the within-phase structure in both phases and the cross-phase structure.
- *Correction (2026-10-08).* The preceding bullet is incorrect. Heterogeneous variances only rescale the correlation matrix, $\Sigma = SRS$ with S the diagonal matrix of SDs, and a positive diagonal rescaling does not affect positive definiteness: the PB block is positive definite whenever its compound-symmetry correlation matrix is (smallest eigenvalue $1 - 0.8 = 0.2$ here). The published compound-symmetry response block is positive definite at the published values (smallest eigenvalue 0.100 for 4 to 32 visits, independent of the schedule), and the failures arise through the biomarker row, where the coupling c_{bm} exceeds the exact ceiling the response block supports (for example, 0.256 to 0.262 at $\rho = 0.8$ without carryover, below both published values 0.3 and 0.6; `docs/36` attributes all 40 failing matrices of its 162-matrix audit to the biomarker row). The failure counts above are unaffected; only the mechanism stated here is wrong (derived; verified numerically). See `analysis/report/compendium-mathematics.tex`, subsections "The joint correlation matrix and the exact ceiling" and "The published compound-symmetry construct", Equations `eq:ce-ceiling` and `eq:ce-cs-decomp`.

OL never fails because expectancy is constant ($\eta_t = 1$) and all variances are homogeneous within each factor block.

3.3 Resolution

Two mechanisms restore PD:

1. **AR(1) autocorrelation replaces compound symmetry.** Under AR(1), nearby timepoints are more correlated than distant ones, which

accommodates variance heterogeneity without forcing a single correlation to cover all pairs. Maximum feasible c_{bm} increases. *Corrected 2026-10-07:* it is 0.256 under CS at $\rho = 0.8$ with no carryover (`docs/36`). At $\rho = 0.7$ it is 0.452 to 0.456 (OL+BDC, Hybrid) under the current `covar` AR(1) structure and 0.474 to 0.480 under the separable AR(1) structure (`docs/34`, `docs/35`). The earlier "0.49 or higher" compared two values of ρ.

2. **Pre-validation via `validateParameterGrid()`.** The revised code tests every parameter combination for PD before simulation begins and reports which cells would require correction. Under the revised parameter grid (`c.bm <= 0.45`, `rho = 0.7`, clinically realistic half-lives), zero of 162 matrices require correction.

Maximum feasible c_{bm} by design:

Table 9: Maximum feasible c_{bm} by design under original and revised parameters (April 2026 grid search)

Design	Original (CS, $\rho = 0.8$)	Revised (AR(1), $\rho = 0.7$)
OL	0.89	0.49
OL+BDC	0.26	0.45
CO	0.25	0.61
N-of-1	0.25	0.53

The publication value of $c_{bm} = 0.6$ exceeds the PD-feasible range for every design except OL. The revised value of $c_{bm} = 0.45$ is within range for all designs.

Note (2026-10-07). The original column agrees with `docs/36` (OL 0.893, OL+BDC 0.262, CO and Hybrid 0.256). The revised column does not agree with the exact closed-form ceilings of `docs/34` at $t_{1/2} = 0$: CO 0.619, N-of-1 (Hybrid) 0.456, OL+BDC 0.452. Under AR(1) without graded coupling, `docs/36` gives OL as 0.576. Prefer the closed-form values. The value 0.45 is inside the range but within 0.006 of the boundary.

3.4 Architecture A Removes the Constraint

Under Architecture A, the BM-BR correlation block is not entered into the covariance matrix (it is applied post-draw as an additive shift). The `c.bm` parameter therefore has no effect on the eigenvalue structure. Testing `c.bm` from 0 to 0.95 in steps of 0.05 across 4 designs and 3 half-lives:

Table 10: Positive definiteness limits on the moderation strength (`c.bm`) for Architecture B and Architecture A

	Architecture B	Architecture A
First PD failure	$c_{bm} = 0.50$	none through $\beta_{bm} = 0.95$
Feasible ceiling	approximately 0.45	unconstrained
Paths failing at 0.60	7 of 9	0 of 9
Paths failing at 0.95	9 of 9	0 of 9

3.5 Residual PD Instrumentation Gap (Finding C1)

The audit-2026-04-16 flagged that all three production implementations still call `make.positive.definite()` silently when a matrix is not PD (root `generateData.R:284-299`; corresponding lines in original, original-extended, tidyverse). No flag is exposed to downstream consumers. The current PD correction rate therefore cannot be verified without instrumentation. The recommended fix is to return a `pd_corrected` flag from `buildSigma()` and aggregate it into per-parameter-set summaries.

Note (2026-10-07, inspected `R/generateData.R`). The repair is still applied without a returned flag. `buildSigma()` now issues a warning with the smallest eigenvalue, but only when `verbose = TRUE`. `docs/34` (Section 1, point 6) measures the perturbation just above the ceiling at no more than 0.004 in any correlation.

The doc 19 (mean moderation implementation notes) also flags that `validateParameterGrid()` builds sigma matrices with the Architecture B BM-BR correlation block regardless of the simulation's actual `dgp_architecture` setting. Under Architecture A with high `c.bm`, the validator may report spurious PD failures. The data generation itself is unaffected because the simulation uses the Architecture A-specific sigma. The warning can be safely ignored when running Architecture A. As of 2026-10-07 `validateParameterGrid()` still takes no `dgp_architecture` argument and calls `buildSigma()` with its default `'mvn'` (inspected).

---

4. Analysis Model: nlme::lme with corCAR1

Source documents: 06-ar1-residual-correlation.tex; 03-audit-and-revision-report.tex; 04-revised-power-analysis.tex.

4.1 Formula Construction

The analysis model adapts to the design. For designs with within-subject drug variation (CO, N-of-1, OL+BDC):

$$Y_{it} = \beta_0 + \beta_1 B_i + \beta_2 t + \beta_D D_{bc,it} + \beta_{bm:D} B_i D_{bc,it} + u_i + \varepsilon_{it}$$

with $D_{bc,it}$ the continuous drug indicator (code `Dbc`; 1 on drug, exponential decay off drug) and $\beta_{bm:D}$ the interaction of interest. The biomarker enters uncentered, as the raw `bm` column (inspected, `R/lme_analysis.R`).

For open-label (no within-subject drug variation):

$$Y_{it} = \beta_0 + \beta_1 B_i + \beta_2 t + \beta_3 B_i \cdot t + u_i + \varepsilon_{it}$$

with β_3 the biomarker-by-time interaction. Random intercept only (`random = ~1|ptID`). Residual correlation `corCAR1(form = ~t|ptID)`.

4.2 Continuous-Time AR(1) Rationale

Trial timepoints are unequally spaced. The N-of-1 design uses weeks `c(4, 8, 9, 10, 11, 12, 16, 20)` with gaps of 1 to 4 weeks. Discrete AR(1) (`corAR1`) assumes equal spacing and is incorrect for these designs. Continuous-time AR(1) (`corCAR1`) handles unequal spacing via $\rho^{|t_1 - t_2|}$ with t in weeks (nlme reports ρ as the parameter Phi). The schedule quoted is the N-of-1 example in `vignettes/01-simulate-trial-data.Rmd` (inspected).

4.3 Why the Random Intercept Alone Is Insufficient

Under compound symmetry the within-person correlation is constant, and a random intercept absorbs it perfectly (residuals independent).

Under AR(1), let the within-participant deviations have total variance σ_T^2 and correlation $\rho^{|t_1 - t_2|}$. A random intercept of variance σ_u^2 leaves residuals of variance $\sigma_\varepsilon^2 = \sigma_T^2 - \sigma_u^2$ and covariance $\sigma_T^2 \rho^{|t_1 - t_2|} - \sigma_u^2$, so

$$\mathrm{Cor}(\varepsilon_{it_1}, \varepsilon_{it_2} \mid u_i) = \frac{\sigma_T^2 \rho^{|t_1 - t_2|} - \sigma_u^2}{\sigma_\varepsilon^2} = \rho^{|t_1 - t_2|} - \left(1 - \rho^{|t_1 - t_2|}\right) \frac{\sigma_u^2}{\sigma_\varepsilon^2}$$

(derived; corrected 2026-10-07). The earlier expression, $\rho^{|t_1 - t_2|} - \sigma_u^2/(\sigma_u^2 + \sigma_\varepsilon^2)$, omitted the rescaling by the residual variance. The qualitative conclusion is unchanged. The random intercept absorbs only part of the correlation. Residual correlation remains positive for nearby timepoints (where $\rho^{|t_1 - t_2|} > \sigma_u^2/\sigma_T^2$) and

is negative for distant ones. `lme4::lmer` treats residuals as independent and therefore underestimates standard errors.

(Qualified 2026-10-08: the direction of the misstatement depends on the contrast and the design, so "underestimates" holds for some designs only. For contiguous on-drug and off-drug blocks the AR(1) contrast variance exceeds the compound-symmetry one (ratio 3.25 at four visits per block), so a random-intercept-only model understates the standard error and the test is anti-conservative, as in the quoted `lmer` crossover sizes of 0.13 to 0.17; for interleaved designs the ratio is smaller. This is argued from the contrast-variance formula, not computed for each design. See `analysis/report/compendium-mathematics.tex`, subsection "What a random intercept absorbs under AR(1)", Equation `eq:inva-ri` and its note.)

4.4 Type I Error Correction

Under AR(1) DGP with `lme4::lmer` analysis (200 replicates, `c.bm = 0`, no censoring, no carryover):

Table 11: Type I error by design and N under lme4::lmer analysis

Design	N=35	N=70
OL	0.13	0.09
OL+BDC	0.09	0.05
CO	0.17	0.13
N-of-1	0.06	0.08

The CO design reaches 17 percent false positives at N=35, 3.4 times nominal. CO and OL are most affected because their drug indicators are most strongly confounded with time-dependent residual structure.

Under the revised `nlme::lme` plus `corCAR1` model:

Table 12: Type I error by design and N under nlme::lme with corCAR1

Design	N=35	N=70
OL	0.03	0.02
OL+BDC	0.03	0.04
CO	0.06	0.08
N-of-1	0.04	0.06

All lie within the central 95 percent binomial range for a true size of 0.05 at 200 replicates, [0.02, 0.08] (computed), with OL at N=70 and CO at N=70 on its

edges. At 1,000 replicates (doc 04), Type I error is 2.5 to 5.8 percent across all designs.

Note (2026-10-07). Later work superseded the conclusion that `nlme::lme` with `corCAR1` and model-based inference is adequate. `docs/45` found real size excess for model-based and Kenward-Roger inference under flexible-time and baseline-row analyses. Its own primary recommendation (an unstructured model with Kenward-Roger) was in turn superseded, because that model rejected at 0.057 pooled and 0.087 at N = 35 and failed every prognostic-biomarker cell. The exploratory stress test of `docs/46` (41 cells, 28,000 trials, 14 analyses, decision rule rerun with all 14) retained ten analyses after the size and sensitivity screens: S, C1, C2, DS, ES, EA, F1, F2, G1 and G2. Of these, ES, EA, F1, G1 and G2 are robust to a prognostic biomarker. G2 and F1 tie on maximin shortfall (0.232 against 0.242), so the rule selects F1, with fewer covariance parameters. F1 is the `corCAR1` strawman with CR2 inference plus an expectancy term (`De + bmc:De`; `docs/46` and `docs/49`). Neither reaches power 0.80 on the grid.

4.5 Error Handling

`lme_analysis()` wraps the fit in a two-level fallback:

1. If `lme` with `corCAR1` fails to converge: fall back to `lme` without correlation structure.
2. If the fallback fails: return NA for all outputs.

Rows with NA outcomes are removed before fitting (nlme uses `na.fail` by default). Audit finding C2 noted that the returned `issingular` column was hard-coded FALSE (`lme_analysis.R:178, 221` at the time); downstream consumers could not distinguish clean fits from singular ones. The recommended fix was to inspect `fit$apVar` for the 'Non-positive-definite approximate variance-covariance' string and check random-effects variance for zero eigenvalues. *Note (2026-10-07, inspected):* `lme_analysis()` now sets `issingular` from `VarCorr(fit)`, TRUE when the random-intercept SD is non-finite or below 10^{-6}, and NA when both fits fail. The `apVar` check was not adopted.

4.6 Coefficient Extraction

nlme returns `summary(fit)$tTable` with columns `Value`, `Std.Error`, `p-value` (contrasted with `lme4::lmer`, which uses `Estimate`, `Std. Error`, `Pr(>|t|)`). The extraction searches the row names for `'bm:Dbc'` or `'Dbc:bm'` and selects the first match:

```
target <- intersect(c('bm:Dbc', 'Dbc:bm'), rownames(c))
if (length(target) == 0) {
```

```
  beta <- betaSE <- p <- as.numeric(NA)
} else {
  target <- target[1]
  p <- c[target, 'p-value']
  beta <- c[target, 'Value']
  betaSE <- c[target, 'Std.Error']
}
```

Audit finding C4 flags this as fragile; `carryover_analysis.R` uses a more flexible pattern with a longer alias list.

---

5. BM-BR Correlation Decay: Derivation from First Principles

Source documents: 08-biomarker-correlation-decay.tex; 09-carryover-correlation-artifact.tex.

5.1 Three Historical Rules

The BM-BR correlation at an off-drug timepoint has been computed three different ways across commits:

1. **Step function** (commit 42ac030, publication): full correlation c_{bm} whenever the adjusted BR mean is nonzero, zero otherwise, $r_t = c_{bm}\,\mathbf{1}[\mu_{BR,t} \neq 0]$.
2. **Proportional scaling** (commit 8609f12, Ron Thomas): $r_t = (\mu_{BR,t}/\mu_{BR,t-1})\,c_{bm}$. Correlation decays proportionally with the ratio of adjacent-timepoint BR means.
3. **Independent exponential** (current HEAD): $r_t = c_{bm}e^{-\lambda t_{sd}}$ (code `lambda_cor`), and $r_t = 0$ before first exposure. Correlation decays exponentially with a parameterized rate. `docs/37` calls this graded coupling, $g_t = \phi_t$.

5.2 Individual-Level Argument

Consider two participants during an on-drug period when the population BR mean is μ_{BR}:

- Participant H with biomarker one SD above mean ($b = +1$).

- Participant L with biomarker one SD below mean ($b = -1$).

From the bivariate normal conditional expectation:

$$E[BR_H] = \mu_{BR} + c_{bm} \cdot \sigma_{BR}$$
$$E[BR_L] = \mu_{BR} - c_{bm} \cdot \sigma_{BR}$$

Biomarker-specific spread: $\Delta_{\text{on}} = 2\, c_{bm}\, \sigma_{BR}$.

5.3 Behavior at Drug Discontinuation

If carryover is pharmacokinetic, drug molecules clear at a rate determined by the half-life, independent of any participant's biomarker. Every participant's drug effect decays by the same factor $\phi_t = (1/2)^{t_{sd}/t_{1/2}}$:

$$E[BR_H^{\text{off}}] = \phi_t(\mu_{BR} + c_{bm}\sigma_{BR}), \quad E[BR_L^{\text{off}}] = \phi_t(\mu_{BR} - c_{bm}\sigma_{BR})$$

Population mean off-drug: $\mu_{BR}^{\text{off}} = \phi_t \mu_{BR}$.

Biomarker-specific spread off-drug: $\Delta_{\text{off}} = \phi_t \Delta_{\text{on}}$. The spread decays by the same factor as the mean.

5.4 Implied Correlation

Setting the implied spread equal to the MVN-derived form $\Delta_{\text{off}} = 2\, r_{\text{off}}\, \sigma_{BR}$ (which holds when σ_{BR} is the same on and off drug; see Section 5.8):

$$\boxed{r_{\text{off}} = \phi_t \cdot c_{bm} = (1/2)^{t_{sd}/t_{1/2}} \cdot c_{bm}}$$

Equivalently in exponential form, since $\phi_t = \exp(-(\ln 2/t_{1/2})\, t_{sd})$, the correlation decay rate (code `lambda_cor`) equals the drug elimination rate:

$$\boxed{\lambda = \frac{\ln 2}{t_{1/2}}}$$

(derived).

5.5 Numerical Validation

N-of-1 Path A, $c_{bm} = 0.5$, $t_{1/2} = 1.0$ week, $\sigma_{BR} = 8$:

Table 13: On-drug and off-drug BR mean and correlation for N-of-1 Path A

Metric	On drug (BD2, week 10)	Off drug (BD3, week 11)
$\mu_{BR,t}$	10.19	5.09
r_t	0.50	0.25
$E[BR \mid b = +1]$	14.19	7.09
$E[BR \mid b = -1]$	6.19	3.09
Spread Δ	8.0	4.0
Spread / mean	0.785	0.786

The spread halves as the mean halves (arithmetic checked 2026-10-07). The ratio of the biomarker-specific spread to the mean is preserved at 0.79. No participant at ± 1 SD has negative expected BR.

5.6 Comparison with Alternative Rules

Off-drug conditional expectations at BD3 ($\mu_{BR,t} = 5.09$, $c_{bm} = 0.5$, $t_{sd} = 1$ week):

Table 14: Off-drug conditional expectations of BR under alternative decay rules

Rule	r_t	$E[BR \mid b = +1]$	$E[BR \mid b = -1]$	Issue
Zero off-drug correlation	0	5.09	5.09	No biomarker signal
Graded, $\lambda = \ln 2 / t_{1/2}$	0.25	7.09	3.09	Consistent
Slow decay, $\lambda = 1$	0.18	6.56	3.62	Under-predicts spread
Full correlation, $r_t = c_{bm}$ (published step rule)	0.50	9.09	1.09	Over-predicts spread

Only graded coupling produces a spread consistent with pharmacokinetic decay. *Corrected 2026-10-07:* the first row was previously labeled "step function". Because $\mu_{BR,t} \neq 0$ at BD3, the published step rule of Section 5.1 assigns full

correlation here, which is the last row. The slow-decay entries were recomputed ($5.09 \pm 0.5e^{-1} \cdot 8$).

5.7 The Publication Artifact

The publication used the step-function rule at $t_{1/2} = 0.1$ and $t_{1/2} = 0.2$ weeks. At these values the carryover residual in the mean one week after discontinuation is at most about 0.1 percent of peak (2^{-10} at $t_{1/2} = 0.1$). The step-function rule assigned the full c_{bm} correlation to these timepoints, producing large biomarker-specific variation around a near-zero mean. Numerical example at BD3 with 0.1-week half-life, $c_{bm} = 0.6$, $\sigma_{BR} = 8$, $\sigma_{bm} = 15.36$:

Table 15: Expected BR by biomarker level under original and revised decay

Biomarker	$E[BR]$ original	$E[BR]$ revised
109 (-1 SD)	-4.8	+0.01
124 (mean)	+0.01	+0.01
140 (+1 SD)	+4.8	+0.01

The ± 1 SD entries are $0.01 \pm 0.6 \cdot 8 = 0.01 \pm 4.8$; the +1 SD entry was previously given as +4.9 (corrected 2026-10-07).

Under the original rule, a low-biomarker participant's conditional expected BR is -4.8 at a timepoint where the drug effect has decayed to about 0.1 percent of peak (previously misstated as 0.01 percent). This is clinically incoherent: it implies that discontinuing the drug worsens symptoms beyond the natural trajectory for low-biomarker participants, specifically because the drug is gone. No rebound effect was hypothesized.

The analysis model sees this spurious biomarker-specific variation as structured noise in the `bm:Dbc` term. It inflates the residual variance of the interaction estimate, widens the standard error of $\widehat{\beta}_{bm:D}$, and reduces the probability of rejecting the null. The effect is largest for designs with many off-drug timepoints (N-of-1 has 4 of 8) and smallest for OL (always on drug).

Under the revised rule, the off-drug correlation at BD3 with a 0.1-week half-life is approximately $0.001\, c_{bm}$, effectively zero. The Hendrickson et al. (2020) finding of a power drop from 0.95 to 0.18 at $t_{1/2} = 0.1$ weeks is therefore primarily a modeling artifact. At clinically realistic half-lives (0.5 to 1.0 weeks), the revised correlation is substantial and carryover does reduce power, but moderately rather than catastrophically.

Note (2026-10-07). The conclusion that the collapse is an artifact of the step rule stands. The mechanism given above (added noise inflating the standard error) has been replaced by an exact account in `docs/36` and `docs/37`, both derived and verified by simulation. Under the step rule the coupling is full at

every off-drug visit with any residual effect, so for every $t_{1/2} > 0$ the on/off *coupling gap* is zero. The interaction slope is then exactly zero, and power falls to the test size. This is a switch at $t_{1/2} = 0$, not a decline. Graded coupling, $g_t = \phi_t = 2^{-t_{sd}/t_{1/2}}$ off drug and 0 before first exposure, removes the collapse (`docs/37`, Section 1, points 2 and 3). The published figure verified in `docs/36` is Hybrid power falling from 0.74 to 0.12 at $N = 70$, $c_{bm} = 0.3$, $t_{1/2} = 0.1$. The "0.95 to 0.18" figure quoted here and in Section 6.3 was not re-verified.

5.8 The Remaining Variance Limitation

The derivation assumes $\sigma_{BR} = 8$ is constant. This is the current implementation. The conditional variance during off-drug periods is nearly unchanged ($\sigma^2_{BR}(1 - r_t^2)$ with small r_t), so participants have large random fluctuations around a decaying mean. This conflates two sources of variation that should be separated:

- **Trait variability**: between-person differences in latent drug responsiveness, existing before drug and persisting regardless of drug status. Should remain constant.
- **State-dependent variability**: variation in acute drug effect, scaling with effect magnitude. Should decay during carryover.

Addressing the limitation would require a time-varying $\sigma_{BR,t}$ or a variance decomposition, which would be substantial restructuring. For the current application of comparing relative power across designs, the fixed-variance assumption affects all designs similarly and is acceptable as a conservative approximation.

---

6. Validated Results

Source documents: 04-revised-power-analysis.tex; 06-ar1-residual-correlation.tex; 14-trial-analysis-implementation-guide.tex.

6.1 Revised Parameter Grid

Table 16: Revised parameter grid with values and rationale

Parameter	Value	Rationale
`dgp_architecture`	`'mvn'` (default); `'mean_moderation'` optional	Dual support
Autocorrelation	AR(1), $\rho = 0.7$	PD-feasible, clinically realistic
c_{bm} (Arch B)	{0, 0.25, 0.45}	Max PD-feasible approximately 0.45 for OL+BDC
β_{bm} (Arch A)	up to 0.95	PD constraint absent
Carryover $t_{1/2}$	{0, 0.5, 1.0} weeks	Clinically realistic
λ (`lambda_cor`)	$\ln 2/t_{1/2}$ (auto)	Pharmacokinetic derivation
Cross-factor (same t)	`c.cf1t` = 0.2	Unchanged from publication
Cross-factor (different t)	`c.cfct` $\times\rho^{\lvert t_i - t_j\rvert}$	AR(1) decay
Analysis model	nlme::lme + corCAR1	Required for Type I control
PD corrections	0 of 162	All matrices valid

6.2 Type I Error at 1,000 Replicates ($c_{bm} = 0$)

Table 17: Type I error by design and N at 1,000 replicates

Design	N=35	N=70
OL	0.029	0.025
OL+BDC	0.030	0.031
CO	0.053	0.058
N-of-1	0.040	0.030

6.3 Power at $c_{bm} = 0.45$, No Censoring, 1,000 Replicates

Table 18: Power by design, N and half-life at $c_{bm} = 0.45$, 1,000 replicates

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	Interpretation
OL	35	0.06	0.06	0.06	Unaffected
OL	70	0.09	0.10	0.09	Unaffected
CO	35	0.19	0.21	0.18	Robust
CO	70	0.36	0.35	0.30	Robust

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	Interpretation
OL+BDC	35	0.31	0.21	0.12	Moderate drop
OL+BDC	70	0.62	0.44	0.21	Large drop
N-of-1	35	0.45	0.33	0.23	Large drop
N-of-1	70	0.75	0.63	0.43	Large drop

Without carryover the design ranking is N-of-1, then OL+BDC, then CO, then OL, at both sample sizes. Under carryover CO draws level with OL+BDC at $t_{1/2} = 0.5$ for N = 35 (0.21 each) and overtakes it at $t_{1/2} = 1.0$ (0.18 against 0.12 at N = 35; 0.30 against 0.21 at N = 70). (Corrected 2026-10-07: the earlier text ranked CO above OL+BDC at low N, contrary to the table.) Carryover sensitivity is moderate and gradual, not the catastrophic drop from 0.95 to 0.18 at $t_{1/2} = 0.1$ reported in the publication (see the note in Section 5.7).

6.4 Runtime

Current implementation runtime on modern hardware (from doc 14):

- 200 replicates: approximately 8 minutes.
- 1,000 replicates: approximately 47 minutes.

Contributing optimizations (from doc 04 appendix): sigma caching (pre-build 162 covariance matrices once), `furrr::future_map` parallel execution across available cores, pre-computed Cholesky factor, vectorized `data.table` outcome construction, pre-allocated output collection, PD pre-validation, automatic core detection. Together these yield roughly a 12x speedup over the publication code.

---

7. Repository Structure and Implementations

Source documents: 01-codebase-overview.tex; audit-2026-04-16.md; and implementations/README.md.

7.1 Four Parallel Implementations

Table 19: The four parallel implementation collections with style, architectures and role

Collection	Style	Architectures	Location	Role
original	data.table	B (MVN) only	`implementations/original/`	Historical baseline
original-extended	data.table	A + B	`implementations/original-extended/`	Production
tidyverse	tidyverse	A + B	`implementations/tidyverse/`	Modern alternative
simple	base R + tidyverse	A only	`implementations/simple/`	Pedagogical sandbox

The installed package at root `R/` is effectively a copy of `original-extended` with additional plotting, carryover analysis, and documentation functions. The seven core R files are byte-identical between root and original-extended except for the `dgp_architecture` threading in `generateData.R` and `generateSimulatedResults.R`.

7.2 Installed Package File Inventory (root R/)

Table 20: Installed package R files with line counts and exports (April 2026)

File	Lines	Exports
buildtrialdesign.R	119	`buildtrialdesign`
censordata.R	89	`censordata`
datadocumentation.R	61	(none; roxygen stubs)
generateData.R	355	`generateData`, `buildSigma`, `validateParameterGrid`
generateSimulatedResults.R	330	`generateSimulatedResults`
lme_analysis.R	237	`lme_analysis`
packagedocumentation.R	19	(package NULL)
plottingfunctions.R	341	`PlotModelingResults`, `plotfactortrajectories`
utilities.R	108	`cumulative`, `modgompertz`, `reknitsimresults`
carryover_analysis.R	377	`characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`

Line counts and export lists are as of April 2026 and were not re-counted. For example, `generateData.R` has since gained the `'combined'` architecture, `moderation_scaling` and `components` arguments (inspected 2026-10-07). NOTATION.md, Part 2, gives the current list of exported functions.

7.3 Where Each Implementation Stands

original: the seven-file data.table code as corrected in the revised repository. Architecture B only; no `dgp_architecture` parameter. Supports OL, CO, OL+BDC, and N-of-1 designs. Has an independent test suite (6 test files, 458 lines) using tinytest naming conventions, driven by a custom `run_tests.R` rather than standard package test infrastructure. Retained as the Architecture B-only reference implementation.

original-extended: adds the `dgp_architecture` argument with `match.arg` enforcement, the Architecture A additive shift block in `generateData()`, and the BM-BR guard in `buildSigma()`. Threads the argument through `generateSimulatedResults()`. Fifty-two additional lines in two files relative to `original`. The `tests/` directory is empty; coverage relies on the root `inst/tinytest/` suite and the parity tests.

tidyverse: a complete reimplementation as a single 991-line `functions.R` with snake_case naming, tidyverse selection helpers, and three forms of carryover decay (exponential, linear, Weibull) exposed via `carryover_decay()`. The `tests/` directory is empty. A non-automated `test-alignment.R` lives alongside `functions.R` and is sourced manually to validate sigma, Gompertz, and data generation against the `original` collection.

simple: a single 577-line `simulation.R` that demonstrates Architecture A mean moderation with a reduced-complexity model (ANCOVA on phase-level means rather than LME plus corCAR1; no sigma caching; no carryover-aware residual structure). Not package- structured. By design, not used for publication-grade power calculations.

7.4 Test Inventory

From the 2026-04-16 audit:

Table 21: Test inventory by location with coverage, assertions and gaps

Location	Files	Coverage	Assertions	Gaps
`inst/tinytest/`	5	cumulative, modgompertz, buildtrialdesign, buildSigma, generateData, lme_analysis	57	No Type I error check; no mean_moderation; no corCAR1 correctness; no PD recovery; no convergence/singular fit test
`implementations/original/tests/`	6	standalone	458 lines	Tinytest-style naming but sourced manually
`implementations/original-extended/tests/`	0	—	0	Empty
`implementations/tidyverse/R/test_alignment.R`	1	Rcpp vs R Gompertz, data parity	5 tinytest	Manual sourcing; no formal framework
`implementations/test-parity-extended-tidyverse.R`	1	extended-tidyverse approximately implementation	144 cells	Parity-only

The root `inst/tinytest/` suite covers core functions but has known coverage gaps flagged as audit finding C7: no test exercises `dgp_architecture = 'mean_moderation'`, null-effect Type I error, `corCAR1` correctness under a known ground-truth ρ, or the silent PD correction path.

Note: the project CLAUDE.md referred to the `testthat` framework at the time. The installed package actually uses `tinytest` (configured as `Suggests: tinytest (>= 1.4.0)` in DESCRIPTION, driven by `tests/tinytest.R`). This discrepancy was flagged in the audit. As of 2026-10-07 the project CLAUDE.md specifies `tinytest` (inspected).

7.5 Bundled Data

- `results_core.rda`: publication Figure 4 results (initial commit, 500 replicates).
- `extracted_bp.rda`: baseline parameters (BP mean/SD, CAPS mean/SD from pilot data).
- `extracted_rp.rda`: response parameters (Gompertz max, disp, rate, sd for tv, pb, br).

- `CTdata.rda`: actual clinical trial data for Vignette 3.
- `results_maxes.rda`, `results_rates.rda`, `results_trajectories.rda`: Figure 5 results.

7.6 Reproducibility Infrastructure

- **Docker**: `rocker/tidyverse:4.5.3` (ARG `R_VERSION=4.5.3`); non-root `analyst` user; `languageserver` for IDE support. Working directory `/home/analyst/project`.
- **renv.lock**: R 4.5.2 (4.4.2 in the CI workflow), approximately 200 packages pinned from Posit PPM.
- **Makefile targets**: `help`, `r`, `rstudio`, `document`, `build`, `check`, `install`, `test`, `vignettes`, `manuscripts`, `deps`, `check-renv`, `check-system-deps`, `docker-build`, `docker-test`, `manuscript-01`, `manuscript-02`.
- **GitHub Actions**: `.github/workflows/r-package.yml` builds in `rocker/tidyverse:4.4.2`, restores from renv.lock, and runs `R CMD check --no-manual --ignore-vignettes` with `rcmdcheck` and `tinytest` installed to a separate CI tool library.
- **Pre-commit hook**: `scripts/pre-commit-parity.sh` runs `implementations/test-parity-original-` to validate cross-implementation parity on each commit.

---

8. Architecture A: Implementation History

Source documents: 19-mean-moderation-implementation-notes.tex; 19-mean-moderation-implementation-notes.md; 02-revision-response-to-reviewer.md.

8.1 Four Attempts

The current Architecture A implementation was reached after three incorrect formulations. Each is recorded because the failure modes illuminate the calibration constraints the correct formulation must satisfy.

Attempt 1: Multiplicative scaling (incorrect).

$$BR_{it}^{\text{mod}} = BR_{it}^{\text{base}}(1 + \beta_{bm} b_i)$$

Problem: during off-drug timepoints BR_{it}^{base} is approximately zero (the Gompertz evaluates to approximately 0 when $t_{od} = 0$). Multiplying near-zero by `(1 + small)` produces near-zero. The signal is proportional to the magnitude

of the base BR, not to the drug exposure. Result: power dropped approximately 60 percent under carryover, indistinguishable from Architecture B. Lesson: multiplicative moderation conflates the interaction signal with the response magnitude.

Attempt 2: Additive with drug exposure decay (incorrect).

$$BR_{it}^{\text{mod}} = BR_{it}^{\text{base}} + \beta_{bm}\, b_i\, h_t\, \sigma_{BR}, \qquad h_t = \begin{cases} 1 & \text{on drug} \\ \phi_t = (1/2)^{t_{sd}/t_{1/2}} & \text{off drug} \\ 0 & \text{never treated} \end{cases}$$

Problem: drug exposure decays during off-drug. At 2.5-week intervals with $t_{1/2} = 1.0$, exposure values are $\{0.18, 0.031, 0.006, 0.001\}$ (computed). Moderation signal at the first off-drug timepoint is $0.45 \cdot 1 \cdot 0.18 \cdot 8 = 0.65$ BR units, falling to 0.004. Against per-timepoint noise $\sigma_{BR} = 8$, the signal is undetectable. Result: power under carryover remained low, comparable to Architecture B. Lesson: tracking drug exposure decay in the moderation signal reintroduces the very phenomenon Architecture A is intended to avoid.

Attempt 3: Additive with time-on-drug and uncentered biomarker (incorrect scaling).

$$BR_{it}^{\text{mod}} = BR_{it}^{\text{base}} + B_i\, t_{od,t}\, \beta_{bm}, \qquad B_i \text{ the raw (uncentered) biomarker}$$

This was the form in the pre-merge 2025 tidyverse codebase (commit 957c8bd, `pm_functions.R` lines approximately 650-670). Problem: the biomarker mean is approximately 124 mmHg (systolic BP) with SD approximately 15; t_{od} reaches 10 to 20 weeks. Moderation shift at an on-drug timepoint with $t_{od} = 10$ is $124 \cdot 10 \cdot 0.45 = 558$ BR units against a BR noise SD of 8, a signal-to-noise ratio of about 70 to 1. Result: power = 1.00 everywhere. Lesson: dimensional analysis would have caught this immediately. The product $B_i t_{od,t}$ has units of mmHg-weeks, which is not commensurate with BR (a symptom-score change).

(Corrected 2026-10-08: the 558 units are the total shift at the mean biomarker, and almost all of it is biomarker-free drift. With $B_i = \mu_{bm} + \sigma_{bm} b_i$ the shift is $\beta_{bm}\mu_{bm}t_{od,t} + \beta_{bm}\sigma_{bm}t_{od,t}\, b_i$; at $\beta_{bm} = 0.45$, $\mu_{bm} = 124.33$, $\sigma_{bm} = 15.36$ and $t_{od,t} = 10$ the drift is 559.5, which the drug and time terms absorb and which carries no interaction information, and the moderation that the interaction term sees is 69.1 per SD of the biomarker, about 8.6 σ_{BR} rather than 70, growing with $t_{od,t}$. The conclusion, a grossly overpowered signal, stands. Verified by direct computation. See `analysis/report/compendium-mathematics.tex`, subsection "Historical mean-moderation formulations and the calibration check", Equation `eq:inva-attempt3`.)

Attempt 4: Centered, on-drug-binary, calibrated (correct).

$$BR_{it}^{\text{mod}} = BR_{it}^{\text{base}} + \beta_{bm}\, b_i\, \mathbf{1}[D_{it} = 1]\, \sigma_{BR}, \qquad b_i = (B_i - \mu_{bm})/\sigma_{bm}$$

Derivation: at $\beta_{bm} = c_{bm}$ this matches Architecture B's conditional expectation from the MVN model,

$$E[BR_{it} \mid B_i, D_{it} = 1] = \mu_{BR} + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_{bm}} \cdot (B_i - \mu_{bm})$$

Why it works:

1. **Centering**: the standardized biomarker b_i has mean zero. The population-average moderation is zero. Only deviations produce differential BR. Correct for an interaction, not a main effect.
2. **Binary indicator**: $\mathbf{1}[D_{it} = 1]$ is 1 during treatment, 0 during off-drug. Unlike t_{od} (grows with exposure) or h_t (decays), it produces a clean constant signal during all on-drug timepoints and zero during off-drug. Carryover affects the BR population means but not the moderation term.
3. **Scaling by** σ_{BR}: a one-SD biomarker shift produces a $\beta_{bm}\sigma_{BR}$ shift in BR. For $\beta_{bm} = 0.45$ and $\sigma_{BR} = 8$ this is 3.6 BR units, calibrated to match the Architecture B conditional expectation.

8.2 Empirical Verification

Drawing 500 participants with identical parameters ($c_{bm} = \beta_{bm} = 0.45$, crossover design, no carryover):

Table 22: Empirical BR shift and residual SD for Architecture A and Architecture B

	Architecture A	Architecture B
BR shift per SD of biomarker	3.36	3.48
Residual SD	5.23	3.89

The shift magnitudes are comparable. Architecture A has slightly higher residual variance because the BM-BR correlation in the covariance matrix is absent (the noise components are independent), whereas Architecture B's MVN draw produces correlated noise that reduces the residual variance in the regression.

Note (2026-10-07). The two shifts lie within sampling error of the theoretical 3.6 at $N = 500$ (SE about 0.2 to 0.3). The residual SDs could not be checked, because the regression behind them is not specified. A regression of a single

BR_{it} on b_i would leave a residual SD of about $\sigma_{BR} = 8$ under Architecture A and $8\sqrt{1 - 0.45^2} = 7.1$ under Architecture B, not 5.23 and 3.89.

Correction (2026-10-08). The preceding note assumed the wrong regression. The `.tex` version of `docs/19` (Sections 2 and 4.1) specifies a regression of the *mean on-drug* BR on the biomarker. For the mean of the four on-drug visits of a crossover path at 2.5-week spacing, with AR(1) correlation $\rho = 0.7$ per week (the `buildSigma()` default), the residual variance is $\sigma^2_{BR}\,\mathbf{1}^\top A_{\text{on}}\mathbf{1}/16$ under Architecture A and that quantity minus $c^2_{bm}\sigma^2_{BR}$ under Architecture B, where A_{on} is the AR(1) correlation block of the on-drug visits. This gives residual SDs of 5.39 and 4.02, and the reported 5.23 and 3.89 lie within one sampling SD of them (sampling SDs about 0.17 and 0.13 at $N = 500$), so the reported values are reproduced (derived; verified). That the source run used this schedule and ρ is inferred from the match, not from the run itself, which was not located. The explanation in the paragraph above the note (correlated noise under Architecture B reduces the residual variance) is consistent with this result. See `analysis/report/compendium-mathematics.tex`, subsection "Historical mean-moderation formulations and the calibration check", Equation `eq:inva-calib`.

8.3 Hendrickson Publication Parameters Under Architecture A

With the original publication grid (`c.bm` up to 0.6, $t_{1/2}$ in $\{0, 0.1, 0.2\}$), Architecture A makes previously infeasible combinations evaluable. Power at $\beta_{bm} = 0.6$, 50 replicates:

Table 23: Architecture A power at $\beta_{bm} = 0.6$ by design, N and half-life

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.1$	$t_{1/2} = 0.2$
N-of-1	70	0.98	0.96	0.96
OL+BDC	70	0.92	0.92	0.92
CO	70	0.66	0.62	0.70
OL	70	0.14	0.16	0.20
N-of-1	35	0.68	0.78	0.70
OL+BDC	35	0.58	0.66	0.68
CO	35	0.40	0.28	0.26

Power at $\beta_{bm} = 0.3$, 50 replicates:

Table 24: Architecture A power at $\beta_{bm} = 0.3$ by design and N without carryover

Design	N	$t_{1/2} = 0$
N-of-1	70	0.48
OL+BDC	70	0.36

Design	N	$t_{1/2} = 0$
CO	70	0.26

Type I error at $\beta_{bm} = 0$ is 0.00 to 0.08 across all designs (nominal 5 percent). Observations:

1. Carryover resilience is consistent with these runs: power shows no systematic trend across the three half-lives for any design under Architecture A. The largest movement, CO at N = 35 (0.40 to 0.26), is about 1.5 standard errors of a difference at 50 replicates. Independence of power from the half-life under on-drug-only mean moderation is derived in `docs/37` (Section 5.3). Note that these half-lives are very short.
2. The design ranking is preserved: N-of-1 > OL+BDC > CO > OL.
3. $\beta_{bm} = 0.6$ is now evaluable. N-of-1 at $N = 70$ achieves 0.96 to 0.98 power, previously unobtainable because the parameter failed PD validation under Architecture B.

---

9. Clinical Application and Trial Analysis

Source document: 14-trial-analysis-implementation-guide.tex.

9.1 Collaborator Deliverables

Three research goals mapped to implemented functions:

Goal 1: Did prazosin improve PTSD symptoms?

```
result <- analyze_trial_extended(td, dat, op, threshold = 5)
print_trial_summary(result)
```

Defined in `R/carryover_analysis.R`. Outputs:

- Main drug effect (`Dbc` coefficient).
- Biomarker-treatment interaction (`bm:Dbc`).
- 95 percent confidence intervals.
- Variance components (random intercept SD, residual SD, and the `corCAR1` correlation ρ, returned as `ar1_phi`).
- Predicted response curves per participant via `predict(fit, newdata)`.

- Clinical decision threshold, returned as `threshold_bm`: `bm_mean + (threshold - beta_Dbc) / beta_interaction`. *Flag (2026-10-07, inspected):* this formula treats the `Dbc` coefficient as the drug effect at the mean biomarker. However, `lme_analysis()` fits the raw, uncentered `bm`, so that coefficient is the effect at $B_i = 0$. The threshold should then be (threshold $- \hat{\beta}_D)/\hat{\beta}_{bm:D}$ without the `bm_mean` offset, or the biomarker should be centered before the fit. The same offset appears in the `predicted_effect` function. This is a code defect in `R/carryover_analysis.R`, left for a code fix.

Status: complete.

Goal 2a: Characterize carryover in actual data.

```
cc <- characterize_carryover(td, dat,
                             half_lives = c(0.25, 0.5, 1.0, 2.0, 4.0))
print_carryover_summary(cc)
# cc$best_t_half, cc$best_model
```

Sweeps candidate half-lives, fits `lme_analysis()` at each, compares AIC/BIC/log-likelihood, returns the best-fitting half-life and full model object. Validated to recover true `t_half = 1.0` from simulated data.

Status: complete.

Goal 2b: Identify best sequence of response measures.

Status: not yet implemented; conceptual only.

Goal 3a: Simulation power calculations.

Status: complete. Validated 1,000-replicate power estimates with Type I error 2.5 to 5.8 percent across designs. Zero PD failures across 162 parameter combinations. Runtime 8 minutes (200 reps), 47 minutes (1,000 reps).

Goal 3b: Actual analysis output structure.

`lme_analysis(..., full_model_out = TRUE)` returns:

- `form`: the model formula.
- `fit`: the full nlme::lme model object.
- `data`: analysis-ready data with derived variables (`Dbc`, `tsd`, `De`).
- `summary`: beta, SE, p-value.

From the full model object, downstream analyses access `summary(fit)$tTable`, `fitted(fit)`, `residuals(fit)`, `ranef(fit)`, `intervals(fit)`, and `VarCorr(fit)`.

9.2 Prazosin-PTSD Context

From docs 02 and 20:

- Resting systolic blood pressure is a proxy for central noradrenergic tone, a PTSD subtype preferentially responsive to prazosin (an alpha-adrenergic antagonist).
- The biomarker is a statistical predictor of response class, not a causal pharmacokinetic mediator. Architecture B is the appropriate DGP.
- Clinical trial history: Raskind et al. 2003, 2007, 2013 (smaller trials, favorable results on nightmares and PTSD symptoms); Raskind et al. 2018 (PACT trial, N = 304, null result on primary endpoint); Mayer et al. 2023 (posttraumatic headache pilot; previously misattributed to Raskind et al.).
- The PACT null result motivates the search for a predictive biomarker and is the central clinical puzzle the pmsimstats framework addresses.

9.3 Implication for Trial Design

From doc 02, Section 5 (precision medicine trial design): the architectural choice matters most in designs that deliberately expose each participant to both treatment states to estimate within-subject responsiveness (crossover, basket, platform, N-of-1 hybrid). Parallel-group enrichment designs escape Architecture B's carryover sensitivity because they have no off-drug phase. Recommendation: when the biological mechanism is uncertain, run the power calculation under both architectures and treat the Architecture B estimate as the conservative anchor for sample size planning. This is consistent with the PATH Statement (Kent et al. 2020; Ann Intern Med 172(1):35-45, doi:10.7326/M18-3667).

---

10. Alternative Analysis Strategies Under Architecture B

Source document: 02-dgp-mean-moderation-vs-mvn.tex, Section 6.

Six candidate strategies for recovering power under Architecture B, evaluated comparatively. The labels 6.1 to 6.7 below are the section numbers of the source document, not of this compendium. Later evaluations of analysis approaches are collected in `docs/43` (catalog) and `docs/46` (stress test).

6.1 Restrict analysis to on-drug observations only.

Eliminates correlation erosion (all retained observations have full `c_bm`). Costs: reduced effective sample size and loss of the off-drug reference, which confounds biological, placebo, and time-varying components. Best for designs with many

on-drug timepoints (OL has 8, Hybrid has 5 to 6). Least useful for CO, which would lose half its observations.

6.2 Weighted analysis.

On-drug observations weight 1. Off-drug observations weight $\exp(-\lambda t_{sd})$. Rationale: information-theoretic. Observations with higher drug exposure carry more information about the biomarker-treatment interaction because the correlation signal is stronger.

6.3 Within-subject contrast.

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC):

$$\bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}} = \omega_0 + \omega_1 \cdot B_i + e_i$$

(symbols renamed 2026-10-07 from α, γ, which are canonical for the significance level and the $BR \times PB$ interaction).

Sidesteps `Dbc` parameterization and the repeated-measures model. Within-subject averaging reduces noise.

6.4 Two-stage random slopes.

Stage 1: fit a random-slopes LME.

$$Y_{it} = \beta_0 + \beta_1 t + \beta_D D_{it} + u_{0i} + u_{1i} D_{it} + \varepsilon_{it}$$

Stage 2: regress the estimated random deviations on the biomarker.

$$\hat{u}_{1i} = \xi_0 + \xi_1 B_i + e_i$$

(renamed from $\delta_0, \delta_1, \eta_i$, which collide with the canonical standardized bias and expectancy covariate).

Separates within-subject treatment effect estimation (affected by carryover) from between-subject biomarker association (which uses cross-sectional relationships).

6.5 Exclusion of contaminated observations.

Exclude the first one or two off-drug timepoints, which carry the most carryover contamination. Later off-drug timepoints provide a cleaner contrast with the washed-out state.

6.6 Design-level solutions.

- Longer washout periods between drug phases.
- More on-drug timepoints relative to off-drug.

- Front-loaded drug exposure (OL+BDC design): a long initial on-drug block establishes the BM-BR correlation before off-drug observations begin.

6.7 Comparative evaluation.

Table 25: Alternative analysis strategies by data discarded, complexity and expected benefit

Strategy	Discards data	Complexity	Expected benefit
On-drug only	Yes	Low	High for OL/Hybrid; poor for CO
Weighted	No	Low	Moderate; depends on weight calibration
Within-subject contrast	Aggregates	Low	Moderate; robust to misspecification
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages
Exclude early off-drug	Some	Low	Moderate; improves contrast clarity
Design modification	N/A	N/A	High; addresses root cause

---

11. Latent-Class and Psychometric Connections

Source document: 02-revision-response-latent-class-expanded.md.

11.1 The Faithful Generative Model

If the biomarker indexes an unobserved responder/non-responder dichotomy, the mechanistically correct DGP is a finite mixture:

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \quad BR_{it} \mid Z_i = z \sim f_z(t, D_{it})$$

with `Z_i` the latent class, `pi(B_i) = Pr(Z_i = 1 | B_i)` typically a logistic function of the biomarker, and class-specific response distributions `f_0, f_1` differing in drug-response parameters.

Key property: carryover under this DGP operates only on the class-specific means and does not erode any correlation signal, because class membership `Z_i` is drug-state-invariant.

11.2 MVN Differential Correlation as Second-Moment Approximation

A two-component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of `(B_i, BR_{it})` whose first two moments match, to leading order, those of a single MVN with `Cor(B, BR) = c_bm` chosen so that `c_bm^2` equals the between-class variance fraction. Under mild regularity conditions (class overlap, moderate `pi`), the mixture's second-moment structure is well-approximated by MVN with differential correlation in treatment states where class-specific means separate. Architecture B as implemented captures the covariance implications of the latent-class mechanism without committing to the discrete generative form.

(Corrected 2026-10-08: `c_bm^2` is not the between-class variance fraction but the *product* of the two between-class variance fractions, $c_{bm}^2 = f_B f_{BR}$ with $f_B = \pi(1-\pi)(\mu_B^1 - \mu_B^0)^2/\mathrm{Var}(B)$ and f_{BR} defined likewise for BR, under within-class independence of B_i and BR_{it}; equivalently $|c_{bm}|$ is their geometric mean. This follows because the covariance is carried by class membership alone, $\mathrm{Cov}(B, BR) = \pi(1-\pi)(\mu_B^1 - \mu_B^0)(\mu_{BR}^1 - \mu_{BR}^0)$ (derived). See `analysis/report/compendium-mathematics.tex`, subsection "The MVN approximation to mixture-induced moderation", Equation `eq:latent-mvn-approx`.)

11.3 Where the Approximation Breaks Down

Three regimes:

1. **Bimodal responses**: when `f_0` and `f_1` are well-separated (non-responders show near-zero drug effect, responders show a large effect), the marginal BR distribution is bimodal and no MVN matches its tails. Power under the true mixture DGP is **higher** than Architecture B predicts because mixture analysis can exploit bimodality.

2. **Strong class-membership gradient**: when `pi(B_i)` is steep (approaching a step function at a threshold), the biomarker behaves almost deterministically as a class label and Architecture A becomes the better approximation.

3. **Class-varying covariance**: if autocorrelation, residual variance, or placebo-response parameters themselves differ between classes, no single-component MVN can represent the generative structure. Analysis models assuming constant covariance will be misspecified.

11.4 Taxonomy of Finite Mixture Models

From the psychometric tradition:

- **Latent class and latent profile analysis** (Lazarsfeld and Henry 1968; Goodman 1974; Clogg 1995; McLachlan and Peel 2000): small number of discrete subpopulations. LCA for categorical indicators; LPA for continuous. R: `poLCA`.
- **Growth mixture models and group-based trajectory models** (Muthen and Shedden 1999; Nagin 1999, 2005): class-specific growth curves with within-class random-effect variation. Closest psychometric analogue of the mixture DGP discussed above. Identifiability hazards (Bauer and Curran 2003; Bauer 2007) apply with full force at N-of-1 sample sizes.
- **Factor mixture models** (Yung 1997; Arminger, Stein, and Wittenberg 1999; Lubke and Muthen 2005): continuous latent factors combined with discrete latent classes. Subsumes both pure continuous (single-class factor model) and pure discrete (LCA with no within-class structure) as limits.
- **Regression mixtures and mixtures of experts** (DeSarbo and Cron 1988; Wedel and DeSarbo 1995; Jacobs et al. 1991): class- specific regression coefficients with covariate-dependent class membership probabilities.

11.5 The Architecture B Spectrum

Architecture B as implemented is the **most restrictive** operational case of a broader family that ranges from mean-only moderation (Architecture A) through combined mean-plus-covariance moderation (Arminger et al. 1999; the canonical Psychometrika treatment) to heterogeneous-random-effects models (Verbeke and Lesaffre 1996; Proust-Lima et al. 2017).

Implication: the substantial power loss under carryover (40 to 60 percent) documented in Section 1.4 is a specific property of the restrictive covariance-only case and may not generalize to mean-plus-covariance or heterogeneous-random-effects variants.

Recommendation for a complete sensitivity analysis: report power under at least three points on the spectrum: Architecture A (mean-only), Architecture B as implemented (covariance-only), and a combined mean-plus-covariance specification of the Arminger form.

11.6 Practical Constraint and Next Extension

Identifiability concerns for mixture-based analysis at N-of-1 trial sample sizes (typically 30 to 100) are acute. Published FMM applications involve hundreds to thousands of participants. The open question is not whether mixture modeling is more faithful (it is), but whether the fidelity gain is recoverable at realistic sample sizes. Analysis-strategy mitigation (Section 10 above) is more likely than mixture-model analysis to deliver recoverable power gains at realistic trial sizes.

If the pmsimstats program pursues a richer DGP, the natural target is the **heterogeneous-random-slopes form** rather than a full FMM. This preserves the analysis model's compatibility with existing N-of-1 estimation machinery (`nlme::lme` with participant-specific random treatment effects), admits a biologically faithful latent- class generative structure at the DGP level, preserves the audit chain of the current framework, and permits direct comparison with the covariance-only baseline under carryover.

R packages relevant in approximate order of fit: `lcmm` (closest tooling for the heterogeneous-random-slopes form), `flexmix` (regression-mixture framework), `mclust`, `gamlss`, `poLCA`, `OpenMx` and `lavaan` with mixture extensions. Reference implementation outside R: Mplus.

---

12. The 2x2 Factorial: DGP Carryover Versus Analysis-Model Carryover

Source document: 10-carryover-analysis-model-assessment.tex.

12.1 The Question and the Framework

Standard methodology (Senn 2002; Jones and Kenward 2014) for assessing carryover in the analysis model is a 2x2 factorial with DGP carryover on one axis and analysis-model carryover on the other, reporting Type I error, power, bias, and confidence interval coverage in each cell. Senn's modern consensus:

- The two-stage procedure (test for carryover first, then condition) is invalid; carryover tests have low power.
- If carryover is scientifically plausible, either design the trial to eliminate it via adequate washout, or always include carryover in the model and accept the associated power loss. Do not test-then-decide.
- For simulation studies: generate data under a range of carryover magnitudes (including zero), analyze under models that include and exclude the carryover term, and report operating characteristics for each combination.

12.2 Three Analysis Approaches

Note (2026-10-07). The approaches below predate the project's current specification labels (NOTATION.md, Part 3). Approach A is E2 (the exposure-weighted predictor $D_{bc,it}$), and Approach C is E1 (binary indicator, carryover ignored). Approach B, a linear time-since-discontinuation covariate, is not E3; E3 adds a lagged just-off-drug indicator. The study proposed in Section 12.3 was carried out, in revised form, as the carryover-sensitivity manuscript, paper 02 (`analysis/report/02-carryover-sensitivity/`).

Approach A: Continuous drug indicator ($D_{bc,it}$, code `Dbc`).

Formula: `Sx ~ bm + t + Dbc + bm:Dbc`. $D_{bc,it}$ equals 1 on drug and $(1/2)^{s\,t_{sd}/t_{1/2}}$ off drug, where s is `op$carryover_scalefactor` (default 1, inspected). Conflates treatment and carryover; captures both treatment-biomarker and carryover- biomarker interactions simultaneously.

Approach B: Separate carryover covariate (tsd).

Formula: `Sx ~ bm + t + Db + bm:Db + tsd`. The linear time-since- discontinuation term absorbs carryover. Closer to standard crossover methodology but uses a linear carryover term, which is misspecified relative to the exponential DGP.

Approach C: Ignore carryover.

Formula: `Sx ~ bm + t + Db + bm:Db`. Off-drug observations treated as if no drug were present. Biases the treatment contrast toward zero, reducing power, and may inflate Type I error if the bias is systematic.

12.3 Proposed Simulation Study

Design:

- DGP factor: $t_{1/2}$ in {0, 0.25, 0.5, 1.0, 2.0} weeks.
- Analysis factor: three models (Ignore, Separate, Continuous).
- Fixed: N = 70, $\rho = 0.7$, all four trial designs.
- Biomarker correlation: c_{bm} in {0, 0.3, 0.45}.
- 1,000 replicates per cell.
- Total cells: $5 \times 3 \times 3 \times 4 = 180$.

Metrics: Type I error (at $c_{bm} = 0$), power, bias, RMSE, coverage.

Expected findings (labels corrected 2026-10-07; they previously numbered the approaches 1 to 3 in the reverse of the A to C order of Section 12.2):

- Approach C (Ignore): correct Type I error at $t_{1/2} = 0$, may inflate at higher carryover; power declines with increasing carryover.

- Approach B (Separate): correct Type I error across all carryover values; power slightly lower at $t_{1/2} = 0$ (cost of extra parameter), higher at $t_{1/2} > 0$; mild bias from linear versus exponential misspecification.
- Approach A (Continuous): correct Type I error if decay rate matches DGP; power depends on whether the conflation helps or hurts.

Infrastructure: models supported via existing codebase through `op$carryover_t1half` and `op$simplecarryover` parameters.

Status (April 2026): proposed simulation study; not yet complete. See the note at the start of Section 12.2 for its later form.

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13. Simplification Alternatives (Not Adopted)

Source document: 17-simplification-plan.tex.

Doc 17 proposed reducing complexity of the pmsimstats framework substantially. The 2026-04-08 merge took a different path (adding Architecture A rather than simplifying). The document's novel contribution is the most thorough formal specification of six analysis methods available in the project, including precision equations and design implications. It is preserved here because the analytical framework remains useful for cross-method comparisons.

13.1 Six Interaction Test Methods

1. ANCOVA on phase-mean change scores (recommended in doc 17).

Collapse to phase means, $\bar{Y}_{i,\text{on}}$ the mean of Y_{it} over visits with $D_{it} = 1$, and similarly $\bar{Y}_{i,\text{off}}$. Compute $\Delta_i = \bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}}$. Regress $\Delta_i = \omega_0 + \omega_1 B_i + e_i$ (renamed from γ_0, γ_1; γ is canonical for the $BR \times PB$ interaction). Test $H_0 \colon \omega_1 = 0$ by a t-test on $\hat{\omega}_1$ with $N - 2$ df. Precision: $\text{Var}(\hat{\omega}_1) \approx \sigma^2_\Delta / (N \,\text{Var}(B))$.

2. Paired t-test on change scores.

Tests `H_0: E[Delta_i] = 0` (main effect, not interaction). Requires dichotomization of the biomarker (e.g., median split) for the interaction. Efficiency loss from dichotomization approximately 36 percent (`2/pi` approximately 0.64 relative efficiency).

3. Generalized Estimating Equations (GEE).

Population-averaged model with sandwich estimator. Key precision parameter: `Var_within(D_i) = D_bar_i (1 - D_bar_i)`, maximized at `D_bar = 0.5` (equal on/off time). For the OL design, `Var_within(D) = 0`, so the interaction is not estimable.

4. Random-intercept LMM.

Subject-specific model. Precision depends on `SS_within(D_i) = sum_t (D_it - D_bar_i)^2`. Design-specific values: OL has 0, OL+BDC approximately 1.5, CO 2.0, N-of-1 approximately 2.9. These values use doc 17's occasion counts: OL+BDC with 8 visits, about 6 on drug; CO with 8 visits, 4 on drug; N-of-1 with 12 visits, about 7 on drug. With binary D and T visits, $SS_{\text{within}} = T\bar{D}(1-\bar{D}) \leq T/4$, so the current 8-visit Hybrid design cannot exceed 2.0 (derived 2026-10-07). In doc 17, L is the block length; K is not defined there and is read from context as the number of blocks. Short blocks (high K, low L) suffer less from autocorrelation than long blocks because fewer consecutive observations are highly correlated.

5. Simple linear regression (ignoring clustering).

Downward-biased standard errors; Type I error inflation with ICC = 0.3 and T = 8 is approximately 15 to 25 percent at nominal alpha = 0.05. Not recommended for inference.

(Corrected 2026-10-08: the inflation applies only to a *between*-participant effect. With ICC 0.3 and $T = 8$ the design effect is $1 + 7 \cdot 0.3 = 3.1$, giving a realized size of $2\Phi(-1.96/\sqrt{3.1}) = 0.266$ (verified). For the within-participant interaction $\beta_{bm:D}$, the project's estimand, OLS charges the contrast the full residual variance $\sigma_u^2 + \sigma_\varepsilon^2$ rather than σ_ε^2, so its standard errors are overstated and the test is conservative, not anti-conservative (derived for a pure within-participant contrast with exchangeable errors; Paper 10 reports a standard-error ratio near 1.3). OLS remains unsuitable for inference, for the opposite reason. See `analysis/report/compendium-mathematics.tex`, subsections "A general decomposition of realized Type I error", Equation `eq:kappa-ols`, and "What a random intercept absorbs under AR(1)".)

6. Repeated-measures ANOVA (biomarker dichotomized).

Requires dichotomization (36 percent information loss). Sphericity assumption violation requires Greenhouse-Geisser or Huynh-Feldt correction, further reducing power. RM-ANOVA retains approximately 0.64 times ANCOVA efficiency. Design-specific behavior: OL uses time as the within-subject factor; CO is a clean two-level factor with balanced phases; N-of-1 with multiple blocks raises sphericity concerns.

13.2 Cross-Design Summary (doc 17, Table 8.1)

Table 26: Interaction term and target of the test across the four designs

	OL	OL+BDC	CO	N-of-1
Interaction term	bm times t	bm times D	bm times D	bm times D

	OL	OL+BDC	CO	N-of-1
Target	Total improvement rate	Drug-specific	Drug-specific	Drug-specific
Confounded with	PB, TV slopes	none	none	none
LMM info	sum bm^2 SS(t)	sum bm^2 (1.5)	sum bm^2 (2.0)	sum bm^2 (2.9)
Transitions	0	1	1	3
Autocorr vulnerability	Low	Medium	Medium	Low per block
Carryover vulnerability	None	Low	Low	High

13.3 The OL Design Is a Fundamentally Different Test

For the open-label design, all six methods face the same problem: there is no within-subject drug variation. The interaction must be formulated as biomarker-by-time. This is a weaker test than the biomarker-by-drug interaction for two reasons: confounding (the slope dY/dt includes BR, PB, and TV; the biomarker-by-time interaction conflates all three) and precision (the within-subject information for the time slope depends on `SS_within(t) = sum(t - t_bar)^2`, which is typically smaller than `SS_within(D)`).

13.4 Adoption Status

Not adopted. The 2026-04-08 merge preserved the LME with corCAR1 analysis model and added Architecture A rather than simplifying to ANCOVA on phase-mean change scores. The doc is retained because the formal specification of analysis method precision across designs is the most detailed in the project and remains useful for power planning in related contexts.

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14. Code Audit Findings (2026-04-16)

Source document: audit-2026-04-16.md.

The audit identified nine issues across three severity tiers.

14.1 P0 (Execution or Correctness Blockers)

Finding T1: Unqualified dplyr calls in the tidyverse Architecture A path.

Location: `implementations/tidyverse/R/functions.R:421-423`.

```
bm_mean <- baseline_param |> filter(cat == 'bm') |> pull(m)
bm_sd   <- baseline_param |> filter(cat == 'bm') |> pull(sd)
br_sd   <- resp_param     |> filter(cat == 'br') |> pull(sd)
```

These lines execute only when `dgp_architecture == 'mean_moderation'`. If called without `library(dplyr)` in scope, they will either fail at runtime or resolve `filter()` to `stats::filter`. The parity test masks the issue because it runs in a tidyverse-loaded session. Fix: qualify both calls with `dplyr::` and add a regression test invoking `generate_data(..., dgp_architecture = 'mean_moderation')` with only declared Imports loaded.

Finding A1: Broken source path.

Location: `analysis/carryover_factorial/02-run-factorial-2025.R:20`.

Sources `analysis/2025/01-pm-functions.R`, which has been moved to an external archive. The script fails at load time. Fix: change the source target to `implementations/tidyverse/R/functions.R` and update downstream function-name references, or restore a copy from the external archive if historical comparison is needed.

14.2 P1 (Publication and CRAN Blockers)

Finding C1: Silent positive-definiteness correction.

Locations: `generateData.R:284-299` (root), `orig/generateData.R:242-253`, `ext/generateData.R:284-299`, `tid/functions.R:337-356`.

When `is.positive.definite(sigma)` returns FALSE, all implementations call `make.positive.definite(sigma, tol = 1e-3)` without recording the correction. Doc 04 reports 24.7 percent of matrices in the publication required correction. The current audit cannot verify the present rate without instrumentation. Fix: return a `pd_corrected` flag from `buildSigma()` and aggregate into per-parameter-set summaries.

Finding C2: Hard-coded `issingular = FALSE`.

Locations: `lme_analysis.R:178, 221` (copies in orig, ext); `tid/functions.R:654`.

The `issingular` column is always FALSE even when the underlying fit is singular. nlme exposes `fit$apVar` and `fit$modelStruct`; singularity is inferable. Fix: detect singularity from `fit$apVar` (character string 'Non-positive- definite approximate variance-covariance') and check random-effects variance for zero

eigenvalues; populate `issingular` accordingly. (Status 2026-10-07: addressed in the root package by a `VarCorr` check; see Section 4.5.)

Finding C5: Unvalidated `carryover_t1half`.

Location: `generateData.R:172-177` (copies).

A negative `carryover_t1half` produces a negative `lambda_cor`, turning `exp(-lambda_cor * tsd)` into an exponentially growing correlation. No validation. Fix: add `stopifnot(carryover_t1half >= 0)` at the top of `generateData()`. *Note (2026-10-07, inspected):* in the current root `buildSigma()` a non-positive half-life sets `lambda_cor` to 0, so the correlation cannot grow. However, the BR carryover mean term `last_on * (1/2)^(tsd / carryover_t1half)` grows without bound for a negative half-life, and there is still no validation.

Finding C6: Stale imports.

The root DESCRIPTION Imports list includes `lme4`, `lmerTest`, `ggpubr`, and `gridExtra`, none of which are called anywhere in `R/`, `vignettes/`, or the three package-style implementations. These are residues from the pre-nlme analysis model and earlier plotting code. Fix: remove them from DESCRIPTION; if any surfaces in a vignette, promote to Suggests. The audit notes these will generate NOTEs on CRAN. (Status 2026-10-07: all four are still listed under Imports in DESCRIPTION; inspected.)

Finding C7: Missing test coverage.

No test exercises `dgp_architecture = 'mean_moderation'`, null-effect Type I error at nominal alpha, `corCAR1` correctness under known `phi` ground truth, or the silent PD correction path. Fix: add four `tinytest` cases (or migrate to `testthat` 3e) covering these gaps.

Finding A2: Missing READMEs in `analysis/` subfolders.

Locations: `analysis/02-carryover-sensitivity/`, `analysis/architecture_comparison/`, `analysis/carryover_factorial/`, `analysis/figure5/`. Each contains numbered driver scripts without a README. Fix: add a one-paragraph README per subfolder describing research question, inputs, outputs, and runtime.

14.3 P2 (Hygiene and Maintainability)

Finding C3: `eval(parse(text = ...))` for column selection.

Locations: `censordata.R:50, 84`; `plottingfunctions.R:136`; `lme_analysis.R:93` (partial). Fix: replace with `get()`, `dat[[name]]`, or `dplyr::all_of(name)`. No user-facing behavior change.

Finding C4: Positional coefficient extraction.

Locations: `lme_analysis.R:193-217` (copies); `tid/functions.R:637-650`; `carryover_analysis.R:71-93`. Relies on `intersect(c('bm:Dbc',`

`'Dbc:bm'), rownames(...))` and `target[1]`. `carryover_analysis` uses a more flexible pattern. Fix: harmonize around the `carryover_analysis` pattern.

14.4 Parity Status

From the audit: `parity_baseline.rds` and the recent `parity_report.rds` differ by two bytes, indicating stable parity between `original-extended` and `tidyverse` at tolerances of 1e-10 (sigma, data) and 1e-6 (beta, SE, p-value). No expected divergences are recorded in `parity_divergences.csv` (zero data rows).

14.5 Publication Readiness Summary

- CRAN blockers: T1 will fail `R CMD check --as-cran`; C6 will generate NOTEs; C7 weakens the submission narrative.
- Manuscript blockers: `A1` will break reviewer reproduction; C1 (silent PD correction) should surface in methods or appendix but the current rate is not instrumented; C2 (singularity flag always FALSE) may affect reported convergence statistics.

The audit judged that, with the P0 and P1 items addressed, the package and manuscript could proceed to submission.

---

15. Parity Development Workflow

Source document: 22-tidyverse-development-parity-workflow.md.

15.1 Baseline and Inner Loop

Before starting a tidyverse refactor:

```
git checkout -b tidyverse-refactor
Rscript implementations/test-parity-extended-tidyverse.R
cp implementations/parity_report.rds implementations/parity_baseline.rds
```

Commit `parity_baseline.rds` with the branch. At the end of development, diff the current `parity_report.rds` against it to confirm only intended differences.

Two inner-loop cadences:

- `--quick` flag: approximately 2 seconds, 16 cells. Use for iteration on a specific function.
- Full sweep: approximately 10 seconds. Run before every commit.

An optional pre-commit hook blocks commits on failure:

```
Rscript implementations/test-parity-extended-tidyverse.R --quick || \
  { echo 'parity broken; run --full for details'; exit 1; }
```

15.2 Edit Classification

- **Pure refactors** (rename, split function, swap dplyr for purrr, extract helper): parity must hold. If it breaks, a bug was introduced.
- **Feature additions with default values**: parity must still hold for default call paths.
- **Behavior changes** (different DGP parameterization, different MVN ordering, different sigma caching): parity will break by design; decide whether to apply to both implementations in lockstep.

Most development falls into the first two categories. The third should be treated as a separate project with explicit scope.

15.3 Breakage Diagnosis

When parity breaks mid-refactor, the test prints the first failing cell. Diagnose in this order:

1. **Sigma failure**: the change likely touched `build_sigma_matrix` or `build_correlation_matrix`. The BM-BR correlation branch is the most common culprit.
2. **Data failure**: touched MVN sampling, post-draw mean moderation, or participant/path assembly.
3. **Analysis failure**: touched formula construction, `Dbc` computation, or `nlme::lme` invocation.

The `--quick` run covers 16 cells in about 2 seconds. Re-running after every approximately 20 lines of changes localizes bugs to the edit that introduced them.

15.4 RNG Guardrails

The parity test calls `set.seed()` at the per-cell level. If a refactor consumes random draws in a different order (for example swapping `MASS::mvrnorm` for a Cholesky-based draw), parity will break even though the code is semantically correct. Defenses:

- Do not change RNG-consuming loops unless `original-extended` is updated in the same commit.
- Keep `n_cores = 1` for parity-relevant comparison. Introduce parallelism only after functional parity is locked in.

15.5 Intentional Divergence

When a refactor intends to change tidyverse behavior in a way that `original-extended` should not follow (for example fixing a bug that lives only in data.table code, or using tidyverse as a testbed for a new feature):

- Note in the commit message or `DIVERGENCES.md` which cells are allowed to fail and why.
- Mark those cells in the script with an `expected_divergent` flag.
- Re-baseline after the intentional divergence lands: `cp parity_report.rds parity_baseline.rds`.

The trap to avoid is letting an intentional divergence mask a subsequent unintentional one.

15.6 Pre-Merge Checklist

- ☐ cp parity_report.rds parity_baseline.rds at start of branch.
- ☐ –quick after every focused edit (approximately 2 s).
- ☐ Full sweep before every commit (approximately 10 s).
- ☐ If parity breaks, read the first failing cell; fix before proceeding.
- ☐ Never change RNG consumption order in a pure refactor.
- ☐ Add edge cells when new features are added.
- ☐ Document any intentional divergence explicitly.

One-liner for diff before merging:

```
a <- readRDS('implementations/parity_baseline.rds')
b <- readRDS('implementations/parity_report.rds')
all.equal(a$grid, b$grid)  # TRUE, or character vector of diffs
```

---

16. Publication Strategy

Source document: 22-journal-target-recommendation.md.

16.1 Manuscript Summary

Manuscript: 'Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover'. The manuscript formalizes the distinction between direct mean moderation (Architecture A) and differential correlation via MVN (Architecture B), with the central finding that architecture choice determines carryover impact on power: 10 to 15 percent loss under Architecture A, 40 to 60 percent under Architecture B. (These magnitudes come from the 50-replicate pilot; see the note in Section 1.4 for the matched-N re-run of paper 01.)

Methodological positioning: intersection of statistical simulation methodology, N-of-1 and crossover trial design, and biomarker- treatment interaction modeling.

16.2 Ranked Journal Targets

1. **Statistics in Medicine** (primary). A leading journal for statistical methodology in clinical trial design. Primary audience (biostatisticians designing trials) would act on the finding. Collection precedent: Senn (2016; Stat Med 35(7):966-977).

2. **Biometrics**. Shi and Ye (2024; Biometrics 80(2):ujae023, previously misattributed to "Yap, Wang, and Harhay") is the closest comparator (carryover impact on power in crossover designs). The manuscript extends this work by showing that the mechanism determining the interaction affects the magnitude of power loss.

3. **Trials**. Highest concentration of N-of-1 methodology papers in the related literature (4 of 50 in the annotated bibliography). Open access, practical orientation. Collection precedents: Haller and Ulm (2018), Hawksworth et al. (2024; Trials 25:263). The latter was previously listed as two papers, "Chen et al. (2024)" and "Senn and Julious (2024)", neither of which could be found; flagged 2026-10-07.

4. **Statistical Methods in Medical Research**. Publishes simulation-based methodological work for trial design. Collection precedent: Senn (2019), doi:10.1177/0962280217726801 (listed originally as "Arends et al. (2019)", a misattributed entry carrying Senn's title; verified 2026-10-07).

5. **Contemporary Clinical Trials Communications**. Practical framing and clinical application fit well. Collection precedent: Sturdevant and Lumley (2021). The originally listed "Liu et al. (2021)" was a misattributed entry for Lim and In (2021), *Korean Journal of Anesthesiology*, doi:10.4097/kja.21165, and is not a precedent for this journal (verified 2026-10-07).

6. **Harvard Data Science Review**. Collection precedent: Schork (2022). More of a perspective venue; heavy simulation results may not fit format.

7. **Healthcare (MDPI)**. Collection precedents: Blackston et al. (2019), Wang and Schork (2019). Fast review; lower stigma risk if strengthened before submission.

Primary recommendation: target **Statistics in Medicine**. Backup: target **Trials** if Statistics in Medicine considers the scope too specialized.

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17. Literature Synthesis

Source document: 20-annotated-bibliography.tex.

Fifty references in the annotated bibliography organized by topic. The summaries below focus on the claims most directly relevant to the pmsimstats framework.

Note (2026-10-07). Each entry below was checked against Crossref and PubMed. Seven were misattributed or had wrong metadata, and they have been corrected in place, with the earlier citation recorded. The same errors probably occur in `docs/20-annotated-bibliography.tex`, which was not checked. DOIs are given for verified entries.

17.1 Core N-of-1 Simulation Methodology

- **Hendrickson, Thomas, Schork, and Raskind (2020)**, Frontiers in Digital Health 2:13, doi:10.3389/fdgth.2020.00013. Simulation study (four designs, LME analysis, 1,000 repetitions) of aggregated N-of-1 designs for detecting predictive biomarker effects. Finding: the N-of-1 (Hybrid) design outperformed OL and OL+BDC but had slightly lower power than the traditional crossover. This is the framework pmsimstats-ng revises. (Corrected 2026-10-07: previously cited as "Thomas et al.", described as deriving analytic power formulae, and summarized as needing fewer participants than parallel-group designs, none of which matches the abstract.)

- **Blackston et al. (2019)**, Healthcare 7(4):137, doi:10.3390/healthcare7040137. 5,000 simulated trials per scenario across four variance structures comparing aggregated N-of-1, parallel RCT, crossover. Finding: aggregated N-of-1 outperforms traditional designs but has elevated Type I error under moderate-to-strong unmodeled carryover.
- **Wang and Schork (2019)**, Healthcare 7(3):84, doi:10.3390/healthcare7030084. Analytical power formulae under AR(1) serial correlation. Finding: serial correlation substantially reduces power; increasing evaluation periods partially compensates.

17.2 Carryover and Temporal Correlation

- **Shi and Ye (2024)**, "Behavioral carry-over effect and power consideration in crossover trials," Biometrics 80(2):ujae023, doi:10.1093/biomtc/ujae023 (corrected 2026-10-07; previously attributed to "Yap, Wang, and Harhay"). Formal framework for behavioral carryover in crossover trials (psychological mechanisms, not pharmacokinetic). Closest comparator for the pmsimstats manuscript. Finding: behavioral carryover biases the basic estimator toward underestimating the treatment effect, costing power. Directly relevant to the prazosin-PTSD application.
- **Sturdevant and Lumley (2021)**, Contemporary Clinical Trials Communications 22:100711, doi:10.1016/j.conctc.2021.100711. Mixed-effects model approach for testing carryover. Finding: 'carryover has been largely ignored except as a nuisance parameter'.
- **Schork (2022)**, Harvard Data Science Review Special Issue 3, doi:10.1162/99608f92.f1eef6f4. Serial correlation in personalized and aggregated studies. Finding: ignoring serial correlation inflates false positives.
- **Tang and Landes (2020)**, PLoS One 15(2):e0228077, doi:10.1371/journal.pone.0228077. Modified t-tests accounting for serial correlation in N-of-1 data. Finding: proposed tests maintain nominal Type I where standard t-tests fail.

17.3 Biomarker-Treatment Interactions

- **Haller and Ulm (2018)**, Trials 19:128, doi:10.1186/s13063-018-2491-0. Simulation study of biomarker-treatment interaction estimation in randomized trials with time-to-event outcomes. Same genre as the pmsimstats manuscript (Architecture A paradigm).
- **Grenet et al. (2020)**, Basic and Clinical Pharmacology and Toxicology 126(1):59-64, doi:10.1111/bcpt.13293. Simulation comparing crossover versus parallel-group for binary predictive biomarker identification. Finding: crossover more efficient when within-subject correlation is high (supports N-of-1 use in pmsimstats).

17.4 Analysis Methods

- **Chen and Chen (2014)**, "A comparison of four methods for the analysis of N-of-1 trials," PLoS One 9(2):e87752, doi:10.1371/journal.pone.0087752 (corrected 2026-10-07; previously attributed to "Senn, Julious, and Araujo"; the Araujo, Julious and Senn paper is a different one, PLoS One 2016, 11(12):e0167167). Four methods for N-of-1 analysis (fixed-effects, random-effects, Bayesian hierarchical, paired). Finding: all similar for population inference; differ substantially for individual treatment effect estimation.
- **Peterlin, Kejzar, and Blagus (2023)**, TEST 32(1):184-210, doi:10.1007/s11749-022-00830-1. Diagnostic tests for correct specification of fixed- and random-effects design matrices in LMM.

17.5 Sample Size

- **Senn (2019)**, Statistical Methods in Medical Research 28(2):372-383, doi:10.1177/0962280217726801. Five distinct criteria for planning sample size in N-of-1 trials. Finding: criterion depends on goal (individual-level, population-level, heterogeneity detection).
- **Schmid and Duan (2014)**. Statistical design and analytic considerations for N-of-1 trials. Chapter 4 in Kravitz RL, Duan N (eds.), *Design and Implementation of N-of-1 Trials: A User's Guide*, AHRQ Publication No. 13(14)-EHC122-EF. Rockville, MD: Agency for Healthcare Research and Quality; 2014. The chapter title and its place in the guide are confirmed; the chapter's author list rests on the AHRQ product page as reported by a web search, because the AHRQ site blocks automated access (2026-10-07). The original entry listed Kravitz as a third author.
- **Hawksworth et al. (2024)**, "A methodological review of randomised n-of-1 trials," Trials 25:263, doi:10.1186/s13063-024-08100-1 (corrected 2026-10-07; previously cited as "Senn, Julious, and Chen (2024), Trials 25:264"). Methodological review of published randomized N-of-1 trials. Finding: widespread inconsistencies in reporting and analytic practice.

17.6 Bayesian Approaches

- **Schmid and Yang (2022)**, "Bayesian models for N-of-1 trials," Harvard Data Science Review, doi:10.1162/99608f92.3f1772ce (corrected 2026-10-07; previously "Schmid and Staudenmayer (2021)"). Review of Bayesian modeling for N-of-1 trials.
- **Daza (2018)**, "Causal analysis of self-tracked time series data using a counterfactual framework for N-of-1 trials," Methods of Information in Medicine 57(1):e10-e21, doi:10.3414/ME16-02-0044 (corrected 2026-10-07; previously cited as Daza, Wac, and Oppezzo, AMIA Annual Symposium

Proceedings 2018:532-541, which was not found). Counterfactual framework for causal inference in N-of-1 trials via self-tracked time series.

17.7 Prazosin-PTSD Clinical Context

- **Raskind et al. (2003)**, American Journal of Psychiatry 160(2):371-373, doi:10.1176/appi.ajp.160.2.371. Initial proof-of-concept (N=10), double-blind crossover; significant reduction in nightmares.
- **Raskind et al. (2013)**, American Journal of Psychiatry 170(9):1003-1010, doi:10.1176/appi.ajp.2013.12081133. Parallel-group RCT (N=67) in active-duty soldiers; significant effects; observed effect sizes and heterogeneity calibrate simulations.
- **Raskind et al. (2018)**, New England Journal of Medicine 378(6):507-517, doi:10.1056/NEJMoa1507598. PACT trial (N=304); no significant difference versus placebo on primary outcome. The null result motivates the pmsimstats framework search for a predictive biomarker.
- **Mayer et al. (2023)**, "Randomized controlled pilot trial of prazosin for prophylaxis of posttraumatic headaches in active-duty service members and veterans," Headache 63(6):751-762, doi:10.1111/head.14529 (corrected 2026-10-07; previously attributed to Raskind et al.). Pilot for posttraumatic headache prophylaxis (N=48; sample size not re-checked).

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18. Core Contribution Statement

The pmsimstats-ng project establishes that the choice between direct mean moderation (Architecture A) and differential correlation (Architecture B) as the DGP for biomarker-treatment interaction in aggregated N-of-1 designs determines whether carryover effects reduce statistical power modestly (approximately 10 to 15 percent, Architecture A) or substantially (approximately 40 to 60 percent, Architecture B). The choice is a claim about biology: Architecture A for biomarkers that determine the magnitude of the drug's biological effect (renal clearance, receptor density, CYP2D6 status); Architecture B for biomarkers that predict response class through shared variance with a drug-responsive physiological component (blood pressure as PTSD subtype marker, baseline severity, inflammatory markers). Most published simulation studies use Architecture A. The Hendrickson et al. (2020) framework, which pmsimstats-ng revises and extends, appears to be unique in the N-of-1 literature in using Architecture B (Section 1.7). Published power estimates for biomarker- moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers operating through an Architecture B mechanism.

The revised implementation embodies seven DGP corrections relative to the publication (AR(1) autocorrelation, exponential BM-BR decay, nlme with cor-

CAR1, scale-factor removal, timepoint-1 guard removal, origin-passing Gompertz, updated defaults), supports both architectures via the `dgp_architecture` parameter, validates Type I error at nominal 5 percent across all designs, and executes 1,000 replicates in 47 minutes with zero silent positive-definiteness corrections.

Note (2026-10-07). This statement is as of April 2026. The power-loss magnitudes are those of the 50-replicate pilot (Section 1.4). The closed-form account of carryover and power is in `docs/37`. The feasible range of c_{bm} is in `docs/34` to `docs/36`, and at the reference value 0.45 it sits within 0.006 of the boundary. The recommended analysis is F1 from the stress test of `docs/46` and `docs/49`, not the plain `corCAR1` model.